

DEPARTMENT OF CHEMISTRY
SIDO KANHU MURMU UNIVERSITY, DUMKA



**Syllabi for the Four-Year Undergraduate
Programme in Chemistry Major, Minor &
Multidisciplinary Courses**

As per the Provisions NEP-2020 (FYUGP) in

**पत्रांक -01/वि0 1-63 / 2021-223 dated-31/01/2023 by the Department of Higher, Technical
Education & Skill Development, Government of Jharkhand.**

**Upgraded & Implemented from 3rd Semester of Academic Session
2022-26 & From 1st Semester of Session 2023-27 Onwards**

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HIGHLIGHTS OF REGULATIONS OF FYUGP AS PER NEP 2020

PROGRAMME DURATION

- The Full-time, Regular UG programme for a regular student shall be for a period of four years with multiple entry and multiple exit options.
- The session shall commence from 1st of July.

ELIGIBILITY

- The selection for admission will be primarily based on availability of seats in the Major subject and marks imposed by the institution. Merit point for selection will be based on marks obtained in Major subject at Class 12 (or equivalent level) or the aggregate marks of Class 12 (or equivalent level) if Marks of the Major subject is not available. Reservation norms of The Government of Jharkhand must be followed as amended in times.
- UG Degree Programmes with Double Major shall be provided only to those students who secure a minimum of overall 75% marks (7.5 CGPA) or higher.
- Other eligibility criteria including those for multiple entry will be in light of the UGC Guidelines for Multiple Entry and Exit in Academic Programmes offered in Higher Education Institutions.

ADMISSION PROCEDURE

- The reservation policy of the Government of Jharkhand shall apply in admission and the benefit of the same shall be given to the candidates belonging to the State of Jharkhand only. The candidates of other states in the reserved category shall be treated as General category candidates. Other relaxations or reservations shall be applicable as per the prevailing guidelines of the University for FYUGP.

VALIDITY OF REGISTRATION

- Validity of a registration for FYUGP will be for maximum for Seven years from the date of registration.

ACADEMIC CALENDAR

- An Academic Calendar will be prepared by the university to maintain uniformity in the CBCS of the UG Honours Programmes, UG Programmes, semesters and courses in the college run under the university (Constituent/Affiliated).
- **Academic Year:** Two consecutive (one odd + one even) semesters constitute one academic year.
- **Semester:** The Odd Semester is scheduled from **July to December** and the Even Semester is from **January to June**. Each week has a minimum of 40 working hours spread over 6 days.
- Each semester will include – Admission, course work, conduct of examination and declaration of results including semester break.
- In order to undergo 8 weeks' summer internship/ apprenticeship during the summer camp, the Academic Calendar may be scheduled for academic activities as below:
 - a) Odd Semester: **From first Monday of August to third Saturday of December**
 - b) Even Semester: **From first Monday of January to third Saturday of May**
- An academic year comprising 180 working days in the least is divided into two semesters, each semester having at least 90 working days. With six working days in a week, this would mean that each semester will have $90/6 = 15$ teaching/ working weeks. Each working week will have 40 hours of instructional time.
- Each year the University shall draw out a calendar of academic and associated activities, which shall be strictly adhered to. The same is non-negotiable. Further, the Department will make all reasonable endeavours to deliver the programmes of study and other

educational services as mentioned in its Information Brochure and website. However, circumstances may change prompting the Department to reserve the right to change the content and delivery of courses, discontinue or combine courses and introduce or withdraw areas of specialization.

PROGRAMME OVERVIEW/ SCHEME OF THE PROGRAMME

- Undergraduate degree programmes of either 3 or 4-year duration, with multiple entries and exit points and re-entry options within this period, with appropriate certifications such as:
 - UG Certificate after completing 1 year (2 semesters) of study in the chosen fields of study provided they complete one vocational course of 4 credits during the summer vacation of the first year or internship/ Apprenticeship in addition to 6 credits from skill-based courses earned during first and second semester.,
 - UG Diploma after 2 years (4 semesters) of study diploma provided they complete one vocational course of 4 credits or internship/ Apprenticeship/ skill based vocational courses offered during first year or second year summer term in addition to 9 credits from skill-based courses earned during first, second, and third semester,
 - Bachelor's Degree after a 3-year (6 semesters) programme of study,
 - Bachelor's Degree (Honours) after a 4-year (8 semesters) programme of study.
 - Bachelor Degree (Honours with Research) after a 4-year (8 semesters) programme of study to the students undertaking 12 credit Research components in fourth year of FYUGP.

CREDIT OF COURSES

The term 'credit' refers to the weightage given to a course, usually in terms of the number of instructional hours per week assigned to it. The workload relating to a course is measured in terms of credit hours. It determines the number of hours of instruction required per week over the duration of a semester (minimum 15 weeks).

- a) One hour of teaching/ lecture or two hours of laboratory /practical work will be assigned per class/interaction.

One credit for Theory = 15 Hours of Teaching i.e., 15 Credit Hours

One credit for Practicum = 30 Hours of Practical work i.e., 30 Credit Hours

- b) For credit determination, instruction is divided into three major components:

Hours (L) – Classroom Hours of one-hour duration.

Tutorials (T) – Special, elaborate instructions on specific topics of one-hour duration

Practical (P) – Laboratory or field exercises in which the student has to do experiments or other practical work of two-hour duration.

CALCULATION OF MARKS FOR THE PURPOSE OF RESULT

- Student's final marks and the result will be based on the marks obtained in Semester Internal Examination and End Semester Examination organized taken together.
- Passing in a subject will depend on the collective marks obtained in Semester internal and End Semester University Examination both. However, students must pass in Theory and Practical Examinations separately.

PROMOTION CRITERIA

First degree programme with single major:

- 1) The Requisite Marks obtained by a student in a particular subject will be the criteria for promotion to the next Semester.
- 2) No student will be detained in odd Semesters (I, III, V & VII).

- 3) To get promotion from Semester-II to Semester-III a student will be required to pass in at least 75% of Courses in an academic year, a student has to pass in minimum 9 papers out of the total 12 papers.
- 4) To get promotion from Semester-IV to Semester-V (taken together of Semester I, II, III & IV) a student has to pass in minimum 18 papers out of the total 24 papers.
- 5) To get promotion from Semester-VI to Semester-VII (taken all together of Semester I, II, III, IV, V & VI) a student has to pass in minimum 26 papers out of the total 34 papers.
- 6) However, it will be necessary to procure pass marks in each of the paper before completion of the course.

First degree programme with dual major:

- 7) Above criteria are applicable as well on the students pursuing dual degree programmes however first-degree programme will remain independent of the performance of the student in dual major courses.
- 8) To get eligible for taking ESE, a student will be required to pass in at least 75% of Courses in an academic year, a student has to pass in minimum 3 papers out of the total 4 papers.
- 9) It will be a necessity to clear all papers of second major programme in second attempt in succeeding session, failing which the provision of dual major will be withdrawn and the student will be entitled for single first-degree programme.

PUBLICATION OF RESULT

- The result of the examination shall be notified by the Controller of Examinations of the University in different newspapers and also on University website.
- If a student is found indulged in any kind of malpractice/ unfair means during examination, the examination taken by the student for the semester will be cancelled. The candidate has to reappear in all the papers of the session with the students of next coming session and his one year will be detained. However, marks secured by the candidate in all previous semesters will remain unaffected.
- There shall be no Supplementary or Re-examination for any subject. Students who have failed in any subject in an even semester may appear in the subsequent even semester examination for clearing the backlog. Similarly, the students who have failed in any subject in an odd semester may appear in the subsequent odd semester examination for clearing the backlog.

Regulation related with any concern not mentioned above shall be guided by the REVISED CURRICULUM AND CREDIT FRAMEWORK FOR THE FOUR-YEAR UNDERGRADUATE PROGRAMMES (FYUGP) As per Provisions of NEP-2020 implemented from the Academic Year 2022-23 onwards for the State Universities of Jharkhand as per पत्रांक -01/वि0 1-63 / 2021-223 dated-31/01/2023 by the Department of Higher, Technical Education & Skill Development, Government of Jharkhand.

COURSE STRUCTURE FOR FYUGP HONOURS/ RESEARCH

Table 1: Credit Framework for Four Year Undergraduate Programme (FYUGP) under State Universities of Jharkhand [Total Credits = 160]

Level of Courses	Semester	MJ; Discipline Specific Courses – Core or Major (80)	MN; Minor from discipline (16)	MN; Minor from vocational (16)	MDC; Multidisciplinary Courses [Life sciences, Physical Sciences, Mathematical and Computer Sciences, Data Analysis, Social Sciences, Humanities, etc.] (9)	AEC; Ability Enhancement Courses (Modern Indian Language and English) (8)	SEC; Skill Enhancement Courses (9)	VAC; Value Added Courses (6)	IAP; Internship/ Dissertation (4)	RC; Research Courses (12)	AMJ; Advanced Courses in lieu of Research (12)	Credits	Double Major (DMJ)
1	2	3	4	5	6	7	8	9	10	11	12	13	14
100-199: Foundation or Introductory Courses	I	4	4		3	2	3	4				20	4+4
	II	4+4		4	3	2	3					20	4+4
Exit Point: Undergraduate Certificate provided with Summer Internship/ Project (4 credits)													
200-299: Intermediate-level Courses	III	4+4	4		3	2	3					20	4+4
	IV	4+4+4		4		2		2				20	4+4
Exit Point: Undergraduate Diploma provided with Summer Internship in 1st or 2nd year/ Project (4 credits)													
300-399: Higher-level Courses	V	4+4+4	4						4			20	4+4
	VI	4+4+4+4		4								20	4+4
Exit Point: Bachelor's Degree													
400-499: Advanced Courses	VII	4+4+4+4	4									20	4+4
	VIII	4		4						12	4+4+4	20	4+4
Exit Point: Bachelor's Degree with Hons. /Hons. with Research												160	224

Note: Honours students not undertaking research will do 3 courses for 12 credits in lieu of a Research project / Dissertation.

Upgraded & Implemented from 3rd Sem. of Session 2022-26 & 1st Sem. of Session 2023-27 Onward

COURSES OF STUDY FOR FOUR YEAR UNDERGRADUATE PROGRAMME 2023 ONWARDS

Table 2: Semester wise Course Code and Credit Points for Single Major:

Semester	Common, Introductory, Major, Minor, Vocational & Internship Courses		Credits
	Code	Papers	
I	AEC-1	Language and Communication Skills (MIL 1 - Hindi/ English)	2
	VAC-1	Value Added Course-1	4
	SEC-1	Skill Enhancement Course-1	3
	MDC-1	Multi-disciplinary Course-1	3
	MN-1A	Minor from Discipline-1	4
	MJ-1	Major paper 1 (Disciplinary/Interdisciplinary Major)	4
II	AEC-2	Language and Communication Skills (MIL 2 - English/ Hindi)	2
	SEC-2	Skill Enhancement Course-2	3
	MDC-2	Multi-disciplinary Course-2	3
	MN-2A	Minor from Vocational Studies/Discipline-2	4
	MJ-2	Major paper 2 (Disciplinary/Interdisciplinary Major)	4
	MJ-3	Major paper 3 (Disciplinary/Interdisciplinary Major)	4
III	AEC-3	Language and Communication Skills (Language Elective 1 - Modern Indian language including TRL)	2
	SEC-3	Skill Enhancement Course-3	3
	MDC-3	Multi-disciplinary Course-3	3
	MN-1B	Minor from Discipline-1	4
	MJ-4	Major paper 4 (Disciplinary/Interdisciplinary Major)	4
	MJ-5	Major paper 5 (Disciplinary/Interdisciplinary Major)	4
IV	AEC-3	Language and Communication Skills (Language Elective - Modern Indian language including TRL)	2
	VAC-2	Value Added Course-2	2
	MJ-6	Major paper 6 (Disciplinary/Interdisciplinary Major)	4
	MJ-7	Major paper 7 (Disciplinary/Interdisciplinary Major)	4
	MJ-8	Major paper 8 (Disciplinary/Interdisciplinary Major)	4
V	MN-1C	Minor from Discipline-1	4
	MJ-9	Major paper 9 (Disciplinary/Interdisciplinary Major)	4
	MJ-10	Major paper 10 (Disciplinary/Interdisciplinary Major)	4
	MJ-11	Major paper 11 (Disciplinary/Interdisciplinary Major)	4
	IAP	Internship/Apprenticeship/Field Work/Dissertation/Project	4
VI	MN-2C	Minor from Vocational Studies/Discipline-2	4
	MJ-12	Major paper 12 (Disciplinary/Interdisciplinary Major)	4
	MJ-13	Major paper 13 (Disciplinary/Interdisciplinary Major)	4
	MJ-14	Major paper 14 (Disciplinary/Interdisciplinary Major)	4
	MJ-15	Major paper 15 (Disciplinary/Interdisciplinary Major)	4
VII	MN-1D	Minor from Discipline-1	4
	MJ-16	Major paper 16 (Disciplinary/Interdisciplinary Major)	4
	MJ-17	Major paper 17 (Disciplinary/Interdisciplinary Major)	4
	MJ-18	Major paper 18 (Disciplinary/Interdisciplinary Major)	4
	MJ-19	Major paper 19 (Disciplinary/Interdisciplinary Major)	4
VIII	MN-2D	Minor from Vocational Studies/Discipline-2	4
	MJ-20	Major paper 20 (Disciplinary/Interdisciplinary Major)	4
	AMJ-1	Advanced Major paper-1 (Disciplinary/Interdisciplinary Major)	4
	AMJ-2	Advanced Major paper-2 (Disciplinary/Interdisciplinary Major)	4
	AMJ-3	Advanced Major paper-3 (Disciplinary/Interdisciplinary Major)	4
For BSc Honours with Research Degree One Requires to Select RC-1 and RC-2 instead of AMJ-1, AMJ-2 & AMJ-3			
VIII	MN-2D	Minor from Vocational Studies/Discipline-2	4
	MJ-20	Major paper 20 (Disciplinary/Interdisciplinary Major)	4
	RC-1	Research Methodology In Chemistry	4
	RC-2	Research Internship/Field Work/Dissertation	8
Total Credits			160

NUMBER OF CREDITS BY TYPE OF COURSE

The hallmark of the new curriculum framework is the flexibility for the students to learn courses of their choice across various branches of undergraduate programmes. This requires that all departments prescribe a certain specified number of credits for each course and common instruction hours (slot time).

Table 3: Overall Course Credit Points for Single Major

Courses	Nature of Courses	3 Year UG Credits	4 Year UG Credits
MAJOR	Core courses	60	80
MINOR	i. Discipline/ Interdisciplinary courses and ii. Vocational Courses	24	32
MULTIDISCIPLINARY	3 Courses	9	9
AEC	Language courses	8	8
SEC	Courses to be developed by the University	9	9
VALUE ADDED COURSES	Understanding India, Environmental Studies, Digital Education, Health & wellness, Summer Internship/ Apprenticeship/ Community outreach activities, etc.	6	6
Internship (In any summer vacation for Exit points or in Semester-V)		4	4
Research/ Dissertation/ Advanced Major Courses	Research Institutions/ 3 Courses		12
Total Credits =		120	160

Table 4: Overall Course Code and Additional Credit Points for Double Major

Courses	Nature of Courses	3 Year UG Credits	4 Year UG Credits
MAJOR 1	Core courses	60	80
MAJOR 2	Core courses	48	64
MINOR	i. Discipline/ Interdisciplinary courses and ii. Vocational Courses	24	32
MULTIDISCIPLINARY	3 Courses	9	9
AEC	Language courses	8	8
SEC	Courses to be developed by the University	9	9
VALUE ADDED COURSES	Understanding India, Environmental Studies, Digital Education, Health & wellness, Summer Internship/ Apprenticeship/ Community outreach activities, etc.	6	6
Internship (In any summer vacation for Exit points or in Semester-V)		4	4
Research/ Dissertation/ Advanced Major Courses	Research Institutions/ 3 Courses		12
Total Credits =		168	224

Table 5: Semester wise Course Code and Additional Credit Points for Double Major:

Semester	Double Major Courses		Credits
	Code	Papers	
I	DMJ-1	Double Major paper-1 (Disciplinary/Interdisciplinary Major)	4
	DMJ-2	Double Major paper-2 (Disciplinary/Interdisciplinary Major)	4
II	DMJ-3	Double Major paper-3 (Disciplinary/Interdisciplinary Major)	4
	DMJ-4	Double Major paper-4 (Disciplinary/Interdisciplinary Major)	4
III	DMJ-5	Double Major paper-5 (Disciplinary/Interdisciplinary Major)	4
	DMJ-6	Double Major paper-6 (Disciplinary/Interdisciplinary Major)	4
IV	DMJ-7	Double Major paper-7 (Disciplinary/Interdisciplinary Major)	4
	DMJ-8	Double Major paper-8 (Disciplinary/Interdisciplinary Major)	4
V	DMJ-9	Double Major paper-9 (Disciplinary/Interdisciplinary Major)	4
	DMJ-10	Double Major paper-10 (Disciplinary/Interdisciplinary Major)	4
VI	DMJ-11	Double Major paper-11 (Disciplinary/Interdisciplinary Major)	4
	DMJ-12	Double Major paper-12 (Disciplinary/Interdisciplinary Major)	4
VII	DMJ-13	Double Major paper-13 (Disciplinary/Interdisciplinary Major)	4
	DMJ-14	Double Major paper-14 (Disciplinary/Interdisciplinary Major)	4
VIII	DMJ-15	Double Major paper-15 (Disciplinary/Interdisciplinary Major)	4
	DMJ-16	Double Major paper-16 (Disciplinary/Interdisciplinary Major)	4
		Total Credits	64

Abbreviations:

AEC: Ability Enhancement Courses
SEC: Skill Enhancement Courses
IAP: Internship/Apprenticeship/ Project
MDC: Multidisciplinary Courses
MJ: Major Disciplinary/Interdisciplinary Courses
DMJ: Double Major Disciplinary/Interdisciplinary Courses
MN: Minor Disciplinary/Interdisciplinary Courses
AMJ: Advanced Major Disciplinary/Interdisciplinary Courses
RC: Research Courses

AIMS OF BACHELOR'S DEGREE PROGRAMME IN CHEMISTRY

The Broad Aims of Bachelor's Degree Programme in Chemistry are intended to provide:

- a. Broad and balance knowledge in chemistry in addition to understanding of key chemical concepts, principles, and theories.
- b. To develop students' ability and skill to acquire expertise over solving both theoretical and applied chemistry problems.
- c. To provide knowledge and skill to the students' thus enabling them to undertake further studies in chemistry in related areas or multidisciplinary areas that can be helpful for self-employment/entrepreneurship.
- d. To provide an environment that ensures cognitive development of students in a holistic manner. A complete dialogue about chemistry, chemical equations and its significance is fostered in this framework, rather than mere theoretical aspects
- e. To provide the latest subject matter, both theoretical as well as practical, such a way to foster their core competency and discovery learning. A chemistry graduates as envisioned in this framework would be sufficiently competent in the field to undertake further discipline-specific studies, as well as to begin domain-related employment.
- f. To mold a responsible citizen who is aware of most basic domain-independent knowledge, including critical thinking and communication.
- g. To enable the graduate, prepare for national as well as international competitive examinations, especially UGC-CSIR NET and UPSC Civil Services Examination.

PROGRAM LEARNING OUTCOMES

The Broad Aims of Bachelor's Degree Programme in Chemistry are:

The student graduating with the Degree B.Sc. (Honours/Research) in Chemistry should be able to understand:

- a) Core competency: Students will acquire core competency in the subject Chemistry, and in allied subject areas.
- b) Systematic and coherent understanding of the fundamental concepts in Physical chemistry, Organic Chemistry, Inorganic Chemistry, Analytical Chemistry, and all other related allied chemistry subjects.
- c) Students will be able to understand use the evidence based comparative chemistry approach to explain the chemical synthesis and analysis.
- d) The students will be able to understand the characterization of materials.
- e) Students will be able to understand the basic principle of equipment, instruments used in the chemistry laboratory.
- f) Students will be able to understand demonstrate the experimental techniques and methods of their area of specialization in Chemistry.
- g) Disciplinary knowledge and skill: A graduate student are expected to be capable of demonstrating comprehensive knowledge and understanding of both theoretical and experimental/applied chemistry knowledge in various fields of interest like Analytical Chemistry, Physical Chemistry, Inorganic Chemistry, Organic Chemistry, Material Chemistry, etc. Further, the student will be capable of using of advanced instruments and related soft-wares for in-depth characterization of materials/chemical analysis and separation technology.
- h) Skilled communicator: The course curriculum incorporates basics and advanced training in order to make a graduate student capable of expressing the subject through technical writing as well as through oral presentation.

- i) Critical thinker and problem solver: The course curriculum also include components that can be helpful to graduate students to develop critical thinking ability by way of solving problems/numerical using basic chemistry knowledge and concepts.
- j) Sense of inquiry: It is expected that the course curriculum will develop an inquisitive characteristic among the students through appropriate questions, planning and reporting experimental investigation.
- k) Team player: The course curriculum has been designed to provide opportunity to act as team player by contributing in laboratory, field-based situation and industry.
- l) Skilled project manager: The course curriculum has been designed in such a manner as to enabling a graduate student to become a skilled project manager by acquiring knowledge about chemistry project management, writing, planning, study of ethical standards and rules and regulations pertaining to scientific project operation
- m) Digitally literate: The course curriculum has been so designed to impart a good working knowledge in understanding and carrying out data analysis, use of library search tools, and use of chemical simulation software and related computational work.
- n) Ethical awareness/reasoning: A graduate student requires to understand and develop ethical awareness/reasoning which the course curriculum adequately provide. Lifelong learner: The course curriculum is designed to inculcate a habit of learning continuously through use of advanced ICT technique and other available techniques/books/journals for personal academic growth as well as for increasing employability opportunity.

**SEMESTER WISE COURSES IN CHEMISTRY MAJOR UPGRADED & IMPLEMENTED FROM 3RD SEM. OF
SESSION 2022-26 & 1ST SEM. OF SESSION 2023-27 ONWARDS**

Table 6: Semester wise Examination Structure in Discipline Courses:

Semester	Courses		Examination Structure			
	Code	Papers	Credits	Mid Semester Theory/ Practical/ Viva (F.M.)	End Semester Theory/ Practical/ Viva (F.M.)	Full Marks
I	MJ-1	MAJOR- 1: ATOMIC STRUCTURE, CHEMICAL BONDING & REDOX REACTIONS	4	25	75	100
II	MJ-2	MAJOR- 2: STATES OF MATTER & CONCEPT OF EQUILIBRIA	4	25	75	100
	MJ-3	MAJOR-3: PRACTICAL-I	4	25	75	100
III	MJ-4	MAJOR-4: GENERAL ORGANIC CHEMISTRY AND HYDROCARBONS	4	25	75	100
	MJ-5	MAJOR-5: PRACTICAL-II	4	25	75	100
IV	MJ-6	MAJOR -6: FUNCTIONAL GROUPS CONTAINING F, Cl, Br & O	4	25	75	100
	MJ-7	MAJOR-7: THERMOCHEMISTRY & CHEMICAL THERMODYNAMICS	4	25	75	100
	MJ-8	MAJOR-8: PRACTICAL-III	4	25	75	100
V	MJ-9	MAJOR-9: ACID, BASES, SALTS, METALLURGY, s & p-BLOCK ELEMENTS	4	25	75	100
	MJ-10	MAJOR-10: FUNCTIONAL GROUPS CONTAINING N & S, HETEROCYCLIC COMPOUNDS, ALKALOIDS, TERPENES	4	25	75	100
	MJ-11	MAJOR-11: PRACTICAL-IV	4	25	75	100
VI	MJ-12	MAJOR-12: d- & f- BLOCK ELEMENTS, COORDINATION CHEMISTRY & NON-AQUEOUS SOLVENTS	4	25	75	100
	MJ-13	MAJOR-13: BIO-ORGANIC CHEMISTRY: AMINO ACIDS, LIPIDS, ENZYMES, NUCLEIC ACIDS & PHARMACEUTICAL COMPOUNDS	4	25	75	100
	MJ-14	MAJOR-14: PHASE EQUILIBRIA, CHEMICAL KINETICS & SURFACE CHEMISTRY	4	25	75	100
	MJ-15	MAJOR-15: PRACTICAL-V	4	25	75	100
VII	MJ-16	MAJOR-16: ORGANIC SPECTROSCOPY, CARBOHYDRATES & DYES	4	25	75	100
	MJ-17	MAJOR-17: ELECTROCHEMISTRY, ELECTRICAL AND MAGNETIC PROPERTIES OF MATERIALS & POLYMERS	4	25	75	100
	MJ-18	MAJOR-18: QUANTUM CHEMISTRY. MOLECULAR SPECTROSCOPY & PHOTOCHEMISTRY	4	25	75	100
	MJ-19	MAJOR-19: PRACTICALS-VI	4	25	75	100
VIII	MJ-20	MAJOR-20: ORGANOMETALLIC AND BIOINORGANIC CHEMISTRY OR SEMINAR & GROUP DISCUSSION (GROUP DISCUSSION-25MARKS, DISSERTATION REPORT -25, PRESENTATION -50 MARKS)	4	25	75	100
	AMJ-1	ADVANCE MAJOR-1: ANALYTICAL CHEMISTRY	4	25	75	100
	AMJ-2	ADVANCE MAJOR-2: GREEN CHEMISTRY & NANO CHEMISTRY	4	25	75	100
	AMJ-3	ADVANCE MAJOR-3: PRACTICALS-VII	4	25	75	100
	Or RC-1	RESEARCH METHODOLOGY	4	25	75	100
	RC-2	PROJECT DISSERTATION/ RESEARCH INTERNSHIP/ FIELD WORK	8	---	---	200
		TOTAL CREDIT	92			

Table 7: Semester wise Course Code and Credit Points for Minor Courses:

Semester	MINOR COURSES		EXAMINATION STRUCTURE			
	Code	Papers	Credits	Mid Semester Theory (F.M.)	End Semester Theory (F.M.)	End Semester Practical/ Viva (F.M.)
I	MN-1A	MINOR-1A: INTRODUCTORY CHEMISTRY	4	15	60	25
III	MN-1B	MINOR-1B: CHEMICAL EQUILIBRIA & FUNCTIONAL GROUPS	4	15	60	25
V	MN-1C	MINOR-1C: CHEMISTRY OF s- & p-BLOCK ELEMENTS AND STATES OF MATTER	4	15	60	25
VII	MN-1D	MINOR-1D: CHEMISTRY OF d- & f-BLOCK ELEMENTS & MOLECULES OF LIFE	4	15	60	25
		TOTAL CREDITS	16			

MAJOR COURSES

SEMESTER-I

MAJOR COURSE

MAJOR COURSE –MJ 1: ATOMIC STRUCTURE, CHEMICAL BONDING & REDOX REACTIONS

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

On completion of this course, the students will be able to understand:

Learning objective:

1. Atomic theory and its evolution.
2. Learning scientific theory of atoms, concept of wave function.
3. Elements in periodic table; physical and chemical characteristics, periodicity.
4. To predict the atomic structure, chemical bonding, and molecular geometry based on accepted models.
5. To understand atomic theory of matter, composition of atom.
6. Identity of given element, relative size, charges of proton, neutron and electrons, and their assembly to form different atoms.
7. Physical and chemical characteristics of elements in various groups and periods according to ionic size, charge, etc. and position in periodic table.
8. Characterize bonding between atoms, molecules, interaction and energetics (ii) hybridization and shapes of atomic, molecular orbitals, bond parameters, bond-distances and energies.
9. Valence bond theory incorporating concepts of hybridization predicting geometry of molecules.
10. Importance of hydrogen bonding, metallic bonding.
11. Principles of Volumetric Analysis.

Course Learning Outcomes:

On successful completion of this course the student should know:

1. Electronic configuration of various elements in periodic table
2. Predicting structure of molecules
3. How hydrogen bonding, metallic bonding is important in common materials' scientific applications to material fabrication

Unit-1 Atomic Structure: (12 Lectures)

Bohr's theory, its limitations and atomic spectrum of hydrogen atom. Wave mechanics: de' Broglie equation, Heisenberg's Uncertainty Principle and its significance, Schrödinger's wave equation, significance of ψ and ψ^2 . Quantum numbers and their significance. Normalized and orthogonal wave functions. Sign of wave functions. Radial and angular wave functions for hydrogen atom. Radial and angular distribution curves. Shapes of s, p, d and f orbitals. Contour boundary and

probability diagrams. Pauli's Exclusion Principle, Hund's rule of maximum multiplicity, Aufbau's principle and its limitations, Variation of orbital energy with atomic number.

Unit-2 Periodicity of Elements: (12 Lectures)

s, p, d, f block elements, the long form of periodic table. Detailed discussion of the following properties of the elements, with reference to s and p-block. Effective nuclear charge, shielding or screening effect, Slater rules, variation of effective nuclear charge in periodic table. , Atomic radii (Vander Waals) ,ionic and crystal radii, Covalent radii (octahedral and tetrahedral) Ionization enthalpy, Successive ionization enthalpies and factors affecting ionization energy. Applications of ionization enthalpy, Electron gain enthalpy, trends of electron gain enthalpy. Electronegativity, Pauling, Mullikan, Allred Rachow scales, electronegativity and bond order, partial charge, hybridization, group electronegativity, Sanderson electron density ratio.

Unit-3 Chemical Bonding: (30 Lectures)

- Ionic bond:** General characteristics, types of ions, size effects, radius ratio rule and its limitations. Packing of ions in crystals. Born-Landé equation with derivation, Madelung constant, expression for lattice energy, Kapustinskii equation. Born-Haber cycle and its application, Solvation energy.
- Covalent bond:** Lewis structure, Valence Shell Electron Pair Repulsion Theory (VSEPR), Shapes of simple molecules and ions containing lone and bond pairs of electrons multiple bonding, sigma and pi-bond approach, Valence Bond theory, (Heitler-London approach). Hybridization containing s, p and s, p, d atomic orbitals, shapes of hybrid orbitals, Bent's rule, Resonance and resonance energy, Molecular orbital theory, Molecular orbital diagrams of simple homonuclear and heteronuclear diatomic molecules: N₂, O₂, C₂, B₂, F₂, CO, NO, and their ions. Covalent character in ionic compounds; polarization, polarizing power and polarizability. Fajan rules. Ionic character in covalent compounds: Bond moment and dipole moment, ionic character from dipole moment and electronegativities.
- Metallic Bond:** Qualitative idea of free electron model, Semiconductors, Insulators.
- Weak Chemical Forces:** Vander Waals, ion-dipole, dipole-dipole, induced dipole dipole-induced dipole interactions, hydrogen bond, effects of hydrogen bonding on melting and boiling points, solubility, dissolution

Unit-4: Oxidation-Reduction and Volumetric Analysis: (6 Lectures)

Redox equations, Balancing by Ion electron method & Oxidation number method.

Disproportionation Reaction. Principles involved in volumetric analysis.

References:

1. Lee, J.D. (2010), Concise Inorganic Chemistry, Wiley India.
2. Huheey, J.E.; Keiter, E.A.; Keiter, R. L.; Medhi, O.K. (2009), Inorganic Chemistry- Principles of Structure and Reactivity, Pearson Education.
3. Douglas, B.E.; McDaniel, D.H.; Alexander, J.J. (1994), Concepts and Models of Inorganic Chemistry, John Wiley & Sons.
4. Atkins, P.W.; Overton, T.L.; Rourke, J.P.; Weller, M.T.; Armstrong, F.A. (2010), Shriver and Atkins Inorganic Chemistry, 5th Edition, Oxford University Press.
5. Pfennig, B. W. (2015), Principles of Inorganic Chemistry. John Wiley & Sons.
6. Housecroft, C. E.; Sharpe, A. G., (2018), Inorganic Chemistry, 5th Edition, Pearson.
7. Wulfsberg, G (2002), Inorganic Chemistry, Viva Books Private Limited.
8. Miessler, G.L.; Fischer P.J.; Tarr, D. A. (2014), Inorganic Chemistry, 5th Edition, Pearson.
9. Shriver, D.; Weller, M.; Overton, T.; Rourke, J.; Armstrong, F. (2014), Inorganic Chemistry, 6th Edition, Freeman & Company
10. Das, A. K.; Das, M. (2014), Fundamental Concepts of Inorganic Chemistry, 1st Edition, Volume CBS Publishers & Distributors Pvt. Ltd.

SEMESTER-II
MAJOR COURSE

MAJOR COURSE- MJ 2: STATES OF MATTER & CONCEPT OF IONIC EQUILIBRIUM

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

On completion of this course, the students will be able to understand:

Learning objective:

1. Familiarization with various states of matter.
2. Physical properties of each state of matter and laws related to describe the states.
3. Calculation of lattice parameters.
4. Electrolytes and electrolytic dissociation, salt hydrolysis and acid-base equilibria.
5. Understanding Kinetic model of gas and its properties.
6. Maxwell distribution, mean-free path, kinetic energies.
7. Behaviour of real gases, its deviation from ideal behaviour, equation of state, isotherm, and law of corresponding states.
8. Liquid state and its physical properties related to temperature and pressure variation.
9. Properties of liquid as solvent for various household and commercial use.
10. Solids, lattice parameters – its calculation, application of symmetry, solid characteristics of simple salts.
11. Ionic equilibria – electrolyte, ionization, dissociation.
12. Salt hydrolysis (acid-base hydrolysis) and its application in chemistry.

Unit-1 Gaseous state: (20 Lectures)

Kinetic Molecular model of a gas: Postulates and derivation of the kinetic gas equation, collision frequency, collision diameter, mean free path and viscosity of gases, their temperature and pressure dependence, relation between mean free path and coefficient of viscosity, calculation of σ from η , variation of viscosity with temperature and pressure. Maxwell distribution and its use in evaluating molecular velocities (average, root mean square and most probable) and average kinetic energy, law of equipartition of energy, degrees of freedom and molecular basis of heat capacities.

Behaviour of real gases: Deviation from ideal gas behaviour, Compressibility factor, Z , Variation of compressibility factor with pressure at constant temperature (plot of Z vs P) for different gases (H_2 , CO_2 , CH_4 and NH_3), Causes of deviation from ideal behaviour. van der Waals equation of state, its derivation and application in explaining real gas behaviour. Boyle's temperature. Isotherms of real gases and their comparison with van der Waals isotherms, continuity of states, critical state, critical and van der Waals constants, law of corresponding states.

Unit-2 Liquid State: (10 Lectures)

Structure and physical properties of liquids; vapour pressure, surface tension, viscosity, and their dependence on temperature, Effect of addition of various solutes on surface tension, cleansing action of detergents. Structure of water.

Unit-3: Solid State: (10 Lectures)

Nature of the solid state, law of constancy of interfacial angles, law of rational indices, Miller indices, elementary ideas of symmetry, symmetry elements and symmetry operations, qualitative idea of point and space groups, seven crystal systems and fourteen Bravais lattices, X-ray diffraction, Bragg's law, a simple account of rotating crystal method and powder pattern method. Analysis of powder diffraction patterns of NaCl, CsCl and KCl. Various types of defects in crystals, Glasses and liquid crystals.

Unit -4 Ionic Equilibrium: (20 Lectures)

Concept of Equilibrium. Le Chatelier's principle and its applications. Relationships between K_p , K_c and K_x for reactions involving ideal gases (Kinetic derivation). Equilibrium between ideal gases and a pure condensed phase.

Strong, moderate and weak electrolytes, degree of ionization, factors affecting degree of ionization, ionization constant and ionic product of water. Ionization of weak acids and bases, pH scale, common ion effect, dissociation constants of mono-, di- and tri-protic acids. Salt hydrolysis, hydrolysis constants, degree of hydrolysis and pH of different salt solutions. Buffer solutions, Henderson equation, buffer capacity, buffer range, buffer action, applications of buffers in analytical chemistry, Solubility and solubility product.

Qualitative treatment of acid–base titration curves (calculation of pH at various stages). Theories of indicators, selection of indicators and their limitations. Multistage equilibria in polyelectrolytes.

Reference Books:

1. Atkins, P. W. & Paula, J. de Atkin's Physical Chemistry 8th Ed., Oxford University Press (2006).
2. Ball, D. W. Physical Chemistry Thomson Press, India (2007).
3. Castellan, G. W. Physical Chemistry 4th Ed. Narosa (2004).
4. Mortimer, R. G. Physical Chemistry 3rd Ed. Elsevier: NOIDA, UP (2009). 5 G. M. Barrow, Tata McGraw Hill (Fifth Edition) (2007)
5. Roy, B. N. Fundamentals of Classical and Statistical Thermodynamics Wiley, 2001 6 Commonly Asked Questions in Thermodynamics. CRC Press, 2011

SEMESTER-II

MAJOR COURSE- MJ 3: MAJOR PRACTICALS-I

Credit: Theory-04, 120 Hours, Full Marks=100, Pass Marks= 40,

Marks: 25 (Attendance=5 + 20 SIE: 1Hr) + 75 (ESE: 6Hrs) = 100, Pass Marks: Practical (SIE + ESE) = 40

Instructions to Question Setter for

Sessional Internal Practical Examination (SIE):

There will be one Sessional Internal Practical Examination of 2 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 20 marks

Attendance=5 Marks

End Semester Examination (ESE):

There will be one Practical Examination of 6 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

Two Experiments = 65 marks

Practical record notebook= 5 marks

Viva-voce= 5 marks

Section -A: Inorganic Chemistry

(A) Acid-Base Titrations

1. Estimation of carbonate and hydroxide present together in mixture.
2. Estimation of carbonate and bicarbonate present together in a mixture.
3. Estimation of free alkali present in different soaps/detergents

(B) Oxidation-Reduction Titrations

1. Estimation of Fe(II) and oxalic acid using standardized KMnO_4 solution.
2. Estimation of oxalic acid and sodium oxalate in a given mixture.
3. Estimation of Fe(II) with $\text{K}_2\text{Cr}_2\text{O}_7$ using internal (diphenylamine, anthranilic acid) and external indicator.

Section -B: Physical Chemistry

Surface Tension Measurements.

1. Determine the surface tension by (i) Drop number (ii) Drop weight method.
2. Study the variation of surface tension of detergent solutions with concentration.
3. Study the effect of the addition of solutes on the surface tension of water at room temperature and explain the observations in terms of molecular interactions:
4. (i)sugar (ii) ethanol (iii) sodium chloride
5. Study the variation of surface tension with different concentration of sodium chloride solutions.

Viscosity measurements using Ostwald's viscometer.

1. Determination of viscosity of aqueous solution of (i) polymer (ii) ethanol and (iii) sugar at room temperature.
2. Viscosity of sucrose solution with the concentration of solute.

Ionic Equilibrium and pH measurements

1. Preparation of buffer solutions of different pH
 - i. Sodium acetate-acetic acid
 - ii. Ammonium chloride-ammonium hydroxide
2. pH metric titration of (i) strong acid vs. strong base, (ii) weak acid vs. strong base.
3. Determination of dissociation constant of a weak acid.
4. Measurement of pH of different solutions like aerated drinks, fruit juices, shampoos and soaps (use dilute solutions of soaps and shampoos to prevent damage to the glass electrode) using pH-meter.

Recommended Books/Reference:

1. Mendham, J., A. I. Vogel's Quantitative Chemical Analysis Sixth Edition, Pearson, 2009.

Upgraded & Implemented from 3rd Sem. of Session 2022-26 & 1st Sem. of Session 2023-27 Onwards

2. Svehala G. and Sivasankar I. B, Vogel's Qualitative Inorganic Analysis, Pearson, India, 2012.
3. Mann, F.G. & Saunders, B.C. Practical Organic Chemistry, Pearson Education (2009)
4. Furniss, B.S.; Hannaford, A.J.; Smith, P.W.G.; Tatchell, A.R. Practical Organic Chemistry, 5thEd., Pearson (2012)
5. Khosla, B.D.; Garg, V.C.; Gulati, A. (2015), Senior Practical Physical Chemistry, S. Chand & Co, New Delhi.
6. Kapoor, K.L. (2019), A Textbook of Physical Chemistry, Vol.7, 1st Edition, McGraw Hill Education.
7. Garland, C. W.; Nibler, J. W.; Shoemaker, D. P. (2003), Experiments in Physical Chemistry, 8th Edition, McGraw-Hill, New York.

SEMESTER-III
MAJOR COURSE

MAJOR COURSE- MJ-4: GENERAL ORGANIC CHEMISTRY AND HYDROCARBONS

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

On completion of this course, the students will be able to understand:

Learning objectives:

1. Basic of organic molecules, structure, bonding, reactivity and reaction mechanisms.
2. Stereochemistry of organic molecules – conformation and configuration, asymmetric molecules and nomenclature.
3. Aromatic compounds and aromaticity, mechanism of aromatic reactions.
4. Understanding hybridization and geometry of atoms, 3-D structure of organic molecules, identifying chiral centres.
5. Reactivity, stability of organic molecules, structure, stereochemistry.
6. Electrophile, nucleophiles, free radicals, electronegativity, resonance, and intermediates along the reaction pathways.
7. Mechanism of organic reactions (effect of nucleophile/leaving group, solvent), substitution vs. elimination.

Learning Outcomes:

1. Design and syntheses of organic molecules.
2. Correlation of Reactivity, stability of organic molecules, structure, stereochemistry.

Unit-1 Basics of Organic Chemistry: (16 Lectures)

Organic Compounds: Classification and Nomenclature, Hybridization, shape of molecules, influence of hybridization on bond properties. Electron Displacement Effects: inductive, electromeric, resonance and mesomeric effects. Tautomerism, hyperconjugation and their applications. Dipole moment, Organic acids and bases, their relative strength. Homolytic and Heterolytic fission with suitable examples. Curly arrow rules, formal charges, Electrophiles and Nucleophiles, Nucleophilicity and basicity, Types, shape and relative stability of reaction intermediates (Carbocations, Carbanions, Free radicals and Carbenes). Aromaticity in benzenoid and non-benzenoid compounds, alternant and non-alternant hydrocarbons, Huckel's rule, annulenes, antiaromaticity, Y-aromaticity, homo-aromaticity, bonding in fullerenes, crown ether complexes and cryptands, inclusion compounds, cyclodextrins, catenanes and rotaxanes. Organic reactions and their mechanism: Addition, Elimination and Substitution reactions.

Unit-2 Stereochemistry: (12 Lectures)

Concept of asymmetry, Fischer Projection, Newmann and Sawhorse projection formulae and their interconversions; Geometrical isomerism: cis–trans and, syn-anti isomerism E/Z notations with C.I.P rules. Optical Isomerism: Optical Activity, Specific Rotation, Chirality/Asymmetry, Enantiomers,

Molecules with two or more chiral-centres, Distereoisomers, meso structures, Racemic mixtures, Relative and absolute configuration: D/L and R/S designations. Threo & Erythro isomers. Cycloalkanes and stability, Baeyer strain theory, Conformation analysis, Energy diagrams of cyclohexane: Chair, Boat and Twist boat forms.

Unit-3 Chemistry of Aliphatic Hydrocarbons: (16 Lectures)

- a) **Alkanes:** Formation of alkanes, Wurtz Reaction, Corey House Synthesis, Kolbe's Synthesis, Free radical substitutions: Halogenation - relative reactivity and selectivity. Lengthening and shortening of carbon chain in alkanes.
- b) **Alkenes and Alkynes:** Formation of alkenes and alkynes by elimination reactions, Mechanism of E1, E2, E1cb reactions. Saytzeff and Hofmann eliminations. Reactions of alkenes: Electrophilic additions their mechanisms (Markownikoff/ Anti Markownikoff addition), mechanism of oxymercuration demercuration, hydroboration-oxidation, ozonolysis, reduction (catalytic and chemical), syn and anti-hydroxylation (oxidation), reaction with NBS, 1, 2- and 1, 4- addition reactions in conjugated dienes and, Diels Alder reaction; Allylic and benzylic bromination and mechanism, e.g. propene, 1-butene, toluene, ethyl benzene. Reactions of alkynes: Acidity, Electrophilic and Nucleophilic additions. Relative reactivity of alkenes and alkynes.

Unit-4 Chemistry of Aromatic Hydrocarbons: (16 Lectures)

- a) **Aromatic Hydrocarbons:** Aromaticity: Aromatic character of arenes, cyclic carbocations/carbanions and heterocyclic compounds with suitable examples. Electrophilic aromatic substitution: halogenation, nitration, sulphonation and Friedel-Craft's alkylation/acylation with their mechanism. Directing effects of substituent groups.
- b) **Polynuclear Hydrocarbons:** Reactions of naphthalene and anthracene: Structure, Preparation and structure elucidation and important derivatives of naphthalene and anthracene.

References:

- 1) Morrison, R. N. & Boyd, R. N. Organic Chemistry, 6th Edn., Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
- 2) Pine S. H. Organic Chemistry, Fifth Edition, McGraw Hill, (2007)
- 3) F. A. Carey, Organic Chemistry, Seventh Edition, Tata McGraw Hill (2008).
- 4) J. Clayden, N. Greeves, S. Warren, Organic Chemistry, 2nd Ed., (2012), Oxford University Press.
- 5) F. A. Carey, R. J. Sundberg, Advanced Organic Chemistry, Part A: Structure and mechanism, Kluwer Academic Publisher, (2000).
- 6) T. W. Graham Solomon: Organic Chemistry, John Wiley and Sons.
- 7) Peter Sykes: A Guide Book to Mechanism in Organic Chemistry, Orient Longman.
- 8) E. L. Eliel: Stereochemistry of Carbon Compounds, Tata McGrawHill.
- 9) L. Finar: Organic Chemistry (Vol. I & II), E. L. B. S.
- 10) R. T. Morrison & R. N. Boyd: Organic Chemistry, Prentice Hall.
- 11) Arun Bahl and B. S. Bahl: Advanced Organic Chemistry, S. Chand
- 12) Ali, Hashmat, Reaction Mechanism in Organic Chemistry, S Chand

SEMESTER-III

MAJOR COURSE- MJ 5: MAJOR PRACTICALS-II

Credit: Theory-04, 120 Hours, Full Marks=100, Pass Marks= 40,

Marks: 25 (Attendance=5 + 20 SIE: 1Hr) + 75 (ESE: 6Hrs) = 100, Pass Marks: Practical (SIE + ESE) = 40

Instructions to Question Setter for

Sessional Internal Practical Examination (SIE):

There will be one Sessional Internal Practical Examination of 2 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 20 marks

Attendance=5 Marks

End Semester Examination (ESE):

There will be one Practical Examination of 6 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

Two Experiments = 65 marks

Practical record notebook= 5 marks

Viva-voce= 5 marks

Section -A: Acquaintance with Chemistry Laboratory

1. Common Laboratory Apparatus

Test tube, Beakers, Erlenmeyer flask, Volumetric flask, graduated cylinder, Pipette, Graduated pipette, Burette, Burette clamp. Funnel, Test tube holder, Bunsen burner, Glass rod, Utility clamp, Spot test plate, Tripod for Bunsen burner, Wash bottle, Spatula, Round-bottom flasks, Glass Condenser, Filter paper, Separatory funnel, Chemical balance, Furnaces etc.

2. Common Symbols of Laboratory Concerns

Biohazard, Highly Flammable, Oxidizing, Corrosive, Harmful/Irritant, Radioactive, Explosive, Toxic, Dangerous for the Environment etc.

3. Common Laboratory Reagents

Common Acids, Common Bases, Common Inorganic/Organic Salts, Organic Compounds, Common Solvents, Difference between Dilute/Concentrated/Fuming liquids.

4. Chemistry Laboratory Techniques

Cutting, Bending & Rounding edge of glass tube & glass rods, fitting glassware's, fitting equipment for Fractional distillation, drawing liquids through pipette, burette & measuring cylinders, Diluting a solution to a known strength, Safe storage of chemicals. Calibration and use of apparatus. Preparation of solutions of different Molarity/Normality of titrants. Use of primary and secondary standard solutions.

Section -B: Organic Chemistry

1. Common Procedures

1. Heating/Boiling with and without condenser, Filtration techniques, Separation techniques, Crystallization techniques.

2. Purification of organic compounds (say naphthalene & others) by crystallization using the following solvents:

a. Water b. Alcohol c. Alcohol-Water d. Acetone e. Hexane f. Toluene

3. Determination of the melting points

a. Determination of the melting points of above compounds and unknown organic compounds
b) (Kjeldahl method and electrically heated melting point apparatus)

a. Effect of impurities on the melting point – mixed melting point of two unknown organic compounds

b. Determination of boiling point of liquid compounds. (Boiling point lower than and more than 100 °C by distillation and capillary method).

SEMESTER-IV
MAJOR COURSE

MAJOR COURSE- MJ 6: FUNCTIONAL GROUPS CONTAINING F, Cl, Br & O

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

:

After completion of the course, the learner shall be able to understand:

Learning objective:

1. Familiarization about classes of organic compounds and their methods of preparation.
2. Basic uses of reaction mechanisms.
3. Name reactions, uses of various reagents and the mechanism of their action.
4. Preparation and uses of various classes of organic compounds.
5. Organometallic compounds and their uses.
6. Organic chemistry reactions and reaction mechanisms.
7. Use of reagents in various organic transformation reactions.

Course Outcomes:

1. Elucidating reaction mechanisms for organic reactions.
2. Use of active methylene groups in organic mechanism and preparation of new organic compounds.

UNIT 1: Chemistry of Halogenated Hydrocarbons: (10 Lectures)

Alkyl halides: Methods of preparation, nucleophilic substitution reactions – SN^1 , SN^2 and SNi mechanisms with stereochemical aspects and effect of solvent etc. Nucleophilic substitution vs. elimination.

Aryl halides: Preparation from diazonium salts. nucleophilic aromatic substitution, $SNAr$, Benzyne mechanism. Relative reactivity of alkyl, allyl/benzyl, vinyl and aryl halides towards nucleophilic substitution reactions.

Organometallic compounds of Mg and Li and their use in synthesis.

UNIT 2: Alcohols, Phenols, Ethers and Epoxides: (10 Lectures)

Alcohols: preparation, properties and relative reactivity of 1° , 2° , 3° - alcohols, Bouveault-Blanc Reduction, Preparation and properties of glycols and glycerol. Pinacol-Pinacolone rearrangement.

Phenols: Preparation and properties, Acidic nature and factors affecting it, Ring substitution reactions, Reimer–Tiemann and Kolbe's–Schmidt Reactions, Fries and Claisen rearrangements with mechanism.

Ethers and Epoxides: Preparation and reaction with acids. Reaction of epoxides with alcohols, ammonia derivatives and $LiAlH_4$

UNIT 3: Carbonyl Compounds: (16 Lectures)

Structure, reactivity and preparation of Carbonyl compounds. Nucleophilic additions, Nucleophilic addition elimination reactions with ammonia derivatives with mechanism. Aldol and Benzoin condensation, Knoevenagel condensation, Claisen-Schmidt, Perkin, Cannizzaro and Wittig reaction, Beckmann and Benzil- Benzilic acid rearrangements, haloform reaction and Baeyer Villiger oxidation, α -substitution reactions, oxidations and reductions (Clemmensen, Wolff- Kishner, LiAlH_4 , NaBH_4 , MPV, PDC and PGC), Addition reactions of unsaturated carbonyl compounds: Michael addition.

UNIT 4: Carboxylic Acids and their Derivatives: (8 Lectures)

Preparation, physical properties and reactions of monocarboxylic acids, Typical reactions of dicarboxylic acids, hydroxy acids and unsaturated acids: succinic/phthalic, lactic, malic, tartaric, citric, maleic and fumaric acids, Preparation and reactions of acid chlorides, anhydrides, esters and amides, Comparative study of nucleophilic substitution at acyl group, Mechanism of acidic and alkaline hydrolysis of esters, Claisen condensation, Dieckmann and Reformatsky reactions, Hofmann bromamide degradation and Curtius rearrangement.

UNIT 5: Chemistry of Active methylene groups: (4 Lectures)

Active methylene compounds: Keto-enol tautomerism. Preparation and synthetic applications of diethyl malonate and ethyl acetoacetate.

Reference Books:

1. P Sykes, A Guide Book to Mechanism in Organic Chemistry, 6th Edition (1997), Orient Longman, New Delhi.
2. Morrison, R. T., Boyd, R. N., Bhattejee, S.K., Organic Chemistry, 7th Edn., Pearson.
3. Acheson, R.M. Introduction to the Chemistry of Heterocyclic compounds, John Welly & Sons (1976).
4. Solomons, T.W., Fryhle Craig, Organic Chemistry, John Wiley & Sons, Inc (2009).
5. McMurry, J.E. Fundamentals of Organic Chemistry, 7th Ed. Cengage Learning India Edition, 2013.
6. Kalsi, P. S. Organic reactions and their mechanisms, New Age Science (2010).
7. Clayden, J., Greeves, N., Warren, S., Wothers, P., Organic Chemistry, Oxford University Press Inc., New York (2001).
8. Reaction mechanism in Organic Chemistry, Hashmat Ali, S. Chand

SEMESTER-IV

MAJOR COURSE- MJ 7: THERMOCHEMISTRY & CHEMICAL THERMODYNAMICS

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

After completion of the course, the learner shall be able to understand:

Learning Objectives:

1. Laws of thermodynamics and concepts.
2. Partial molar quantities and its attributes.
3. Dilute solution and its properties.
4. Understanding the concept of system, variables, heat, work, and laws of thermodynamics.
5. Understanding the concept of heat of reactions and use of equations in calculations of bond energy, enthalpy, etc.
6. Understanding the concept of entropy; reversible, irreversible processes. Calculation of entropy using 3rd law of thermodynamics.
7. Understanding the application of thermodynamics: Joule Thompson effects, partial molar quantities.
8. Understanding theories/thermodynamics of dilute solutions

Course Learning Outcomes:

On successful completion of this course the student should know the:

1. Use of thermochemical equations for calculation of energy and related terms.
2. Use of thermodynamics in explaining chemical behaviour of solute/solvent and reactions.
3. Study of calorimeter principle and its use.

UNIT 1: Introduction & First Law of thermodynamics: (8 Lectures)

Intensive and extensive properties, thermodynamic variables, state and path functions, isolated, closed and open systems, reversible, irreversible and cyclic processes. Zeroth law of thermodynamics. First law of Thermodynamics: Concept of heat, q, work, w, internal energy, enthalpy, relation between heat capacities, calculations of q, w, U and H for reversible and irreversible processes. Expression for work done under free expansion of gases for isothermal and adiabatic conditions.

UNIT 2: Thermochemistry: (9 Lectures)

Heat of reactions: standard states, enthalpy of formation of molecules and ions. Enthalpy of reactions (combustion, neutralization, solution etc) and its applications, calculation of bond energy, bond dissociation energy and resonance energy from thermochemical data, effect of temperature (Kirchhoff's equations) and pressure on enthalpy of reactions.

UNIT 3: Second & Third Law of Thermodynamics: (7 Lectures)

Concept of entropy, thermodynamic scale of temperature, statement of the second law of thermodynamics, molecular and statistical interpretation of entropy. Calculation of entropy change for reversible and irreversible processes.

Third Law: Statement of third law, concept of residual entropy, calculation of absolute entropy of molecules.

UNIT 4: Free Energy Functions: (6 Lectures)

Gibbs and Helmholtz energy, variation of S , G , A with T , V , P , Free energy change and spontaneity. Relation between Joule-Thomson coefficient and other thermodynamic parameters, inversion temperature, Gibbs Helmholtz equation, Maxwell relations, thermodynamic equations of state.

UNIT 5: Partial molar quantities: (8 Lectures)

Partial molar quantities, dependence of thermodynamic parameters on composition, Gibbs-Duhem equation, chemical potential of ideal mixtures, change in thermodynamic functions in mixing of ideal gases.

UNIT 6: Dilute solutions: (10 Lectures)

Dilute solutions, lowering of vapour pressure, Raoult's and Henry's Laws and their applications. Colligative properties of solutions, abnormal colligative properties, Van't Hoff's factor. Thermodynamic derivation using chemical potential to derive relations between the (i) relative lowering of vapour pressure, (ii) elevation of boiling point, (iii) Depression of freezing point, (iv) osmotic pressure and amount of solute. Applications in calculating molar masses of normal, dissociated and associated solutes in solution. Azeotropes.

UNIT 7: Thermodynamics of Chemical Equilibrium (12 Lectures)

Criteria of thermodynamic equilibrium, degree of advancement of reaction, chemical equilibria in ideal gases, concept of fugacity. Thermodynamic derivation of relation between Gibbs free energy of reaction and reaction quotient. Coupling of exoergic and endoergic reactions. Equilibrium constants and their quantitative dependence on temperature, pressure and concentration. Free energy of mixing and spontaneity; thermodynamic derivation of relations between the various equilibrium constants K_p , K_c and K_x .

Reference Books:

1. Peter, A. & Paula, J. de. Physical Chemistry 9th Ed., Oxford University Press (2011).
2. Castellan, G. W. Physical Chemistry 4th Ed., Narosa (2004).
3. Engel, T. & Reid, P. Physical Chemistry 3rd Ed., Prentice-Hall (2012).
4. Assael, M. J.; Goodwin, A. R. H.; Stamatoudis, M.; Wakeham, W. A. & Will, S. Commonly Asked Questions in Thermodynamics. CRC Press: NY (2011).
5. Laidler K. J. and Meiser J. M. Physical Chemistry Third Edition (International) 1999
6. Levine I. N., Physical Chemistry, Fourth Edition, McGraw-Hill (International), 1995.
7. McQuarrie D. A. and Simon J. D. Physical Chemistry- A Molecular Approach, University Science Books, 1998.

SEMESTER-IV

MAJOR COURSE- MJ 8: MAJOR PRACTICALS-III

Credit: Theory-04, 120 Hours, Full Marks=100, Pass Marks= 40,

Marks: 25 (Attendance=5 + 20 SIE: 1Hr) + 75 (ESE: 6Hrs) = 100, Pass Marks: Practical (SIE + ESE) = 40

Instructions to Question Setter for

Sessional Internal Practical Examination (SIE):

There will be one Sessional Internal Practical Examination of 2 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 20 marks

Attendance=5 Marks

End Semester Examination (ESE):

There will be one Practical Examination of 6 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

Two Experiments = 65 marks

Practical record notebook= 5 marks

Viva-voce= 5 marks

I. Organic Chemistry

1. Detection of extra elements in organic compounds.
2. Functional group test for nitro, amine and amide groups
3. Functional group tests for alcohols, phenols, carbonyl and carboxylic acid group.
4. Qualitative analysis of unknown organic compounds containing simple functional groups (alcohols, carboxylic acids, phenols and carbonyl compounds)

5. Organic preparations:

- a. Oxidation of Benzaldehyde to benzoic acid.
- b. Hydrolysis of amides and esters.
- c. Preparation of Semi carbazone derivatives of the following compounds: acetone, ethyl methyl ketone, cyclohexanone, benzaldehyde.
- d. Preparation of methyl orange.

II. Thermochemistry

- (a) Determination of heat capacity of a calorimeter for different volumes using change of enthalpy data of a known system (method of back calculation of heat capacity of calorimeter from known enthalpy of solution or enthalpy of neutralization).
- (b) Determination of heat capacity of the calorimeter and enthalpy of neutralization of hydrochloric acid with sodium hydroxide.
- (c) Calculation of the enthalpy of ionization of ethanoic acid.
- (d) Determination of heat capacity of the calorimeter and integral enthalpy (endothermic and exothermic) solution of salts.
- (e) Determination of basicity/proticity of a polyprotic acid by the thermochemical method in terms of the changes of temperatures observed in the graph of temperature versus time for different additions of a base. Also calculate the enthalpy of neutralization of the first step.
- (f) Determination of enthalpy of hydration of copper sulphate.
- (g) Study of the solubility of benzoic acid in water and determination of ΔH .

Reference Books:

1. Mann, F.G. & Saunders, B.C. Practical Organic Chemistry, Pearson Education (2009)
2. Furniss, B.S.; Hannaford, A.J.; Smith, P.W.G.; Tatchell, A.R. Practical Organic Chemistry, 5th Ed., Pearson (2012)
3. Khosla, B.D.; Garg, V. C. & Gulati, A., Senior Practical Physical Chemistry, R. Chand & Co.: New Delhi (2011).
4. Athawale, V. D. & Mathur, P. Experimental Physical Chemistry New Age International: New Delhi (2001).

Upgraded & Implemented from 3rd Sem. of Session 2022-26 & 1st Sem. of Session 2023-27 Onwards

SEMESTER-V
MAJOR COURSE

**MAJOR COURSE- MJ 9: ACID, BASES, SALTS, METALLURGY, s & p-BLOCK ELEMENTS,
INORGANIC POLYMERS**

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

After completion of the course, the learner shall be able to understand:

Learning Objectives: -

On completion of this course, the students will be able to understand:

1. Acids, Bases and salts and various concepts of acid and base.
2. General Principles of Metallurgy
3. Chemistry of s and p-block elements.
4. Chemistry of noble gases.
5. Redox reactions in hydrometallurgy processes.
6. Structure, bonding of s and p block materials and their oxides/compounds.
7. Chemistry of boron compounds and their structures.
8. Chemistry of noble gases and their compounds; application of VSEPR theory in explaining structure and bonding.

Course Learning Outcomes:

On successful completion of this course the student should know:

1. Extraction of metals through metallurgical operations and their uses.
2. Bonding of various s and p block elements.
3. Use of boron compounds.
4. Applications of acid-base concepts.

UNIT 1: Acids, Bases and Salts: (15 Lectures)

Arrhenius concept of acid and base and its limitations, Bronsted-Lowry concept of acid and base and its limitations, Solvated proton, Relative strengths of acids and bases, Levelling effect and levelling solvents, Types of acid-base reactions, Salts and their classifications, Lewis concept of acid and base and its limitations, Classification of Lewis acids and bases into hard and soft categories, Hard and Soft Acid Base (HSAB) principle and its implications, Theoretical basis of hardness and softness, Electronegativity and hardness and softness, Acid -base strength and hardness and softness, Lux-Flood concept of acid and bases.

UNIT-2: General Principle of Metallurgy: (12 Lectures)

Standard Electrode Potential and its application to inorganic reactions. Occurrence of metals based on standard electrode potentials. Latimer diagrams and Frost diagrams and their applications, Ellingham diagrams for reduction of metal oxides using carbon or carbon

monoxide as reducing agent. Electrolytic Reduction, Hydrometallurgy. Methods of purification of metals: Electrolytic Kroll process, Parting process, van Arkel- de Boer process and Mond's process, Zone refining.

Unit-3: Chemistry of s and p Block Elements: (17 Lectures)

Inert pair effect, Relative stability of different oxidation states, diagonal relationship and anomalous behaviour of first member of each group. Allotropy and catenation. Complex formation tendency of s and p block elements. Hydrides and their classification ionic, covalent and interstitial. Basic beryllium acetate and nitrate.

Structure, bonding, preparation, properties and uses. Boric acid and borates, boron nitrides, borohydrides (diborane) carboranes and graphitic compounds, silanes, Oxides and oxoacids of nitrogen, Phosphorus and chlorine. Per-oxo acids of Sulphur inter-halogen compounds, poly-halide ions, pseudo-halogens, properties of halogens.

UNIT-4: Noble Gases: (8 Lectures)

Occurrence and uses, rationalization of inertness of noble gases, Clathrates; preparation and properties of XeF_2 , XeF_4 and XeF_6 , Bonding in noble gas compounds (Valence bond and MO treatment for XeF_2), Shape of noble gas compounds (VSEPR theory).

UNIT-5: Inorganic Polymers: (8 Lectures)

Introduction, Types of inorganic polymers, comparison with organic polymers, synthesis, structural aspects and applications of Silicones and Siloxanes, Borazines, Phosphazenes and Silicates.

Reference Books:

1. Lee, J.D. Concise Inorganic Chemistry, ELBS, 1991.
2. Douglas, B.E; Mc Daniel, D.H. & Alexander, J.J. Concepts & Models of Inorganic Chemistry 3rd Ed., John Wiley Sons, N.Y. 1994.
3. Greenwood, N.N., Earnshaw. Chemistry of the Elements, Butterworth-Heinemann. 1997.
4. Cotton, F.A. & Wilkinson, G. Advanced Inorganic Chemistry, Wiley, VCH, 1999.
5. Rodger, G.E. Inorganic and Solid State Chemistry, Cengage Learning India Edition, 2002.
6. Miessler, G. L. & Donald, A. Tarr. Inorganic Chemistry Fourth Ed., Pearson, 2010
7. Atkins, P. W and Shriver D. N. Atkins' Inorganic Chemistry 5th Ed. Oxford University Press (2010).

SEMESTER-V

MAJOR COURSE- MJ 10: FUNCTIONAL GROUPS CONTAINING N & S, HETEROCYCLIC COMPOUNDS, ALKALOIDS, TERPENES

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

After completion of the course, the learner shall be able to understand:

Learning Objectives: -

1. Nitrogen containing functional groups and their reactions.
2. Familiarization with polynuclear hydrocarbons and their reactions.
3. Heterocyclic compounds and their reactions.
4. Alkaloids and Terpenes
5. Understanding reactions and reaction mechanism of nitrogen containing functional groups.
6. Understanding the reactions and mechanisms of diazonium compounds.
7. Understanding the structure and their mechanism of reactions of selected polynuclear hydrocarbons.
8. Understanding the structure, mechanism of reactions of selected heterocyclic compounds.
9. Classification, structure, mechanism of reactions of few selected alkaloids and terpenes.

Course outcome:

1. Use of benzene diazonium salt in organic synthesis.
2. Applications of heterocyclic compounds in pharmaceuticals/drugs and the mechanism of actions.
3. Pharmaceuticals/Biomedical applications of alkaloids and terpenes.
4. Nitrogen containing organic compounds/heterocyclic compounds in synthetic chemistry.

UNIT-1: Nitrogen Containing Functional Groups (20 Lectures)

Preparation and important reactions of nitro and compounds, nitriles and isonitriles Amines: Effect of substituent and solvent on basicity; Preparation and properties: Gabriel phthalimide synthesis, Carbylamine reaction, Mannich reaction, Hoffmann's exhaustive methylation, Hofmann- elimination reaction; Distinction between 1°, 2° and 3° amines with Hinsberg reagent and nitrous acid. Diazonium salts: Preparation and synthetic applications.

UNIT-2: Sulphur Containing Compounds: (6 Lectures)

Preparation and reactions of thiols, thioethers including 1,3-dithiane and sulphonic acids

UNIT-3: Heterocyclic Compounds: (20 Lectures)

Classification and nomenclature, Structure, aromaticity in 5-numbered and 6-membered rings containing one heteroatom; Synthesis, reactions and mechanism of substitution reactions of Furan, Pyrrole (Paal-Knorr synthesis, Knorr pyrrole synthesis, Hantzsch synthesis), Thiophene, Pyridine (Hantzsch synthesis), Pyrimidine, Structure elucidation of indole, Fischer indole synthesis and Madelung synthesis), Structure elucidation of quinoline and isoquinoline, Skraup synthesis, Friedlander's synthesis, Knorr quinoline synthesis, Doebner-Miller synthesis, Bischler- Napieralski reaction, Pictet-Spengler reaction, Pomeranz-Fritsch reaction Derivatives of furan: Furfural and

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furoic acid.

UNIT-4: Alkaloids (8 Lectures)

Natural occurrence, General structural features, Isolation and their physiological action
Hoffmann's exhaustive methylation, Emde's modification, Structure elucidation and synthesis of
Hygrine and Nicotine. Medicinal importance of Nicotine, Hygrine, Quinine, Morphine, Cocaine,
and Reserpine.

UNIT-5: Terpenes (6 Lectures)

Occurrence, classification, isoprene rule; Elucidation of structure and synthesis of Citral, Neral
and α -terpineol.

Recommended Text Books/references:

1. Morrison, R. T., Boyd, R. N., Bhatteejee, S.K., Organic Chemistry, 7th Edn., Pearson.
2. Acheson, R.M. Introduction to the Chemistry of Heterocyclic compounds, John Wiley & Sons (1976).
3. Solomons, T.W., Fryhle Craig, Organic Chemistry, John Wiley & Sons, Inc (2009).
4. McMurry, J.E. Fundamentals of Organic Chemistry, 7th Ed. Cengage Learning India Edition, 2013.
5. Kalsi, P. S. Organic reactions and their mechanisms, New Age Science (2010).
6. Clayden, J.; Greeves, N.; Warren, S.; Wothers, P.; Organic Chemistry, Oxford University Press Inc., New York (2001).
7. Singh, J.; Ali, S.M. & Singh, J. Natural Product Chemistry, Prajati Parakashan (2010).
8. Bansal R. K. Heterocyclic Chemistry: Syntheses, Reactions and Mechanisms, New Age, Third Edition (1999).
9. Ali, Hashmat, Reaction Mechanism in Organic Chemistry, S Chand

SEMESTER-V

MAJOR COURSE- MJ 11: MAJOR PRACTICALS-IV

Credit: Theory-04, 120 Hours, Full Marks=100, Pass Marks= 40,

Marks: 25 (Attendance=5 + 20 SIE: 1Hr) + 75 (ESE: 6Hrs) = 100, Pass Marks: Practical (SIE + ESE) = 40

Instructions to Question Setter for

Sessional Internal Practical Examination (SIE):

There will be one Sessional Internal Practical Examination of 2 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 20 marks

Attendance=5 Marks

End Semester Examination (ESE):

There will be one Practical Examination of 6 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

Two Experiments = 65 marks

Practical record notebook= 5 marks

Viva-voce= 5 marks

I. Gravimetric Analysis:

- Estimation of Nickel (II) using Dimethylglyoxime (DMG).
- Estimation of Barium as BaSO_4
- Estimation of Magnesium in pyrolusite
- Estimation of Iron in Fe_2O_3 by precipitating iron as $\text{Fe}(\text{OH})_3$.

II. Organic Chemistry

- Functional group test for nitro, amine and amide groups
- Qualitative analysis of unknown organic compounds containing simple functional groups (nitro, amide, nitriles and isonitriles, amines)

3. Organic preparations:

- Acetylation of one of the following compounds: amines (aniline, o-, m-, p-toluidines and o-, m-p-anisidine) and phenols (β -naphthol, vanillin, salicylic acid) by any one method: (Using conventional method and Using green chemistry approach).
- Benzoylation of one of the amines (aniline, o-, m-, p- toluidines and o-, m-, p- anisidine) and one of the phenols (β -naphthol, resorcinol, p-cresol) by Schotten-Baumann reaction.
- Oxidation of ethanol/ isopropanol (Iodoform reaction).
- Bromination (any one)
 - Acetanilide by conventional methods.
 - Acetanilide using green approach (Bromate-bromide method)
- Nitration: (any one)
 - Acetanilide/nitrobenzene by conventional method.
 - Salicylic acid by green approach (using ceric ammonium nitrate).
- Selective reduction of meta dinitrobenzene to m-nitroaniline.
- Reduction of p-nitrobenzaldehyde by sodium borohydride.
- Aldol condensation with either conventional or green method.
- Benzil-Benzilic acid rearrangement.

Collected solid samples may be used for recrystallization, melting point and TLC.

Reference Books

- J. A. Kent: Riegel's Handbook of Industrial Chemistry, CBS Publishers, New Delhi.
- S. S. Dara: A Textbook of Engineering Chemistry, S. Chand & Company Ltd. New Delhi.
- A. K. De, Environmental Chemistry: New Age International Pvt., Ltd, New Delhi.
- S. M. Khopkar, Environmental Pollution Analysis: New Age Int. Publisher, New Delhi. Mann, F.G. & Saunders, B.C. Practical Organic Chemistry, Pearson Education (2009)

5. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R. Practical Organic Chemistry, 5th Ed., Pearson (2012)
6. Garland, C. W.; Nibler, J. W. & Shoemaker, D. P. Experiments in Physical Chemistry 8th Ed.; McGraw-Hill: New York (2003).
7. Halpern, A. M. & McBane, G. C. Experimental Physical Chemistry 3rd Ed.; W.H. Freeman & Co.: New York (2003).

SEMESTER-VI

MAJOR COURSE- MJ 12: d- & f- BLOCK ELEMENTS, COORDINATION CHEMISTRY & NON-AQUEOUS SOLVENTS

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

After completion of the course, the learner shall be able to understand:

Learning objective:

1. Coordination compounds – its nomenclature, theories, d-orbital splitting in complexes, chelate.
2. Transition metals, its stability, color, oxidation states and complexes.
3. Lanthanides, Actinides – separation, color, spectra and magnetic behavior
4. Bioinorganic chemistry – metal ions in biological system, its toxicity; hemoglobin.
5. Understanding the nomenclature of coordination compounds/complexes, Molecular orbital theory, d-orbital splitting in tetrahedral, octahedral, square planar complexes, chelate effects.
6. Understanding the transition metals stability in reactions, origin of colour and magnetic properties.
7. Understanding the separation of Lanthanoids and Actinoids, its color, spectra and magnetic behavior.
8. The non-aqueous solvents and their role in chemical reaction.

Course outcomes:

1. IUPAC nomenclature of coordination compounds/complexes.
2. Prediction of structure of complexes using various theories; color and magnetic properties of different complexes.
3. Use of lanthanide/actinide compounds in industries.
4. Different types of non-aqueous solvents and different types of chemical reactions taking place in them.

UNIT-1: d-Block Elements: Transition Elements: (15 Lectures)

General group trends with special reference to electronic configuration, colour, variable valency, magnetic and catalytic properties, and ability to form complexes. Stability of various oxidation states and e.m.f. (Latimer & Bsworth diagrams). Difference between the first, second and third transition series. Oxidation states displayed by Cr, Fe, Co, Ni and Cu. A study of the following compounds (including preparation and important properties); Peroxo compounds of chromium, $K_2Cr_2O_7$, $KMnO_4$, $K_4[Fe(CN)_6]$, sodium nitroprusside, $[Co(NH_3)_6]Cl_3$, $Na_3[Co(NO_2)_6]$.

UNIT-2: f-Block elements: Lanthanoids and Actinides: (10 Lectures)

Electronic configuration, oxidation states, color, spectra and magnetic behavior, lanthanide contraction, separation of lanthanides (ion-exchange method only).

UNIT-3: Coordination Chemistry: (25 Lectures)

Werner's theory, EAN rule, piano-stool compounds, valence bond theory (inner and outer orbital complexes), Crystal field theory, d-orbital splitting, weak and strong fields, pairing energies, factors affecting the magnitude of (Δ). Octahedral vs. tetrahedral coordination, tetragonal distortions from octahedral geometry Jahn-Teller theorem, square planar complexes, d orbital splitting in trigonal bipyramidal, square pyramidal and cubic ligand field environments, CFSE, Variation of lattice energies, enthalpies of hydration and crystal radii variations in halides of first and second row transition metal series, Qualitative aspect of Ligand field theory, MO diagrams of representative coronation complexes, IUPAC nomenclature of coordination compounds, isomerism in coordination compounds. Stereochemistry of complexes with the coordination number 4 and 6, Chelate effect.

UNIT-4: Non-Aqueous Solvents:(10 Lectures)

Solvents and their role during chemical reactions, Classification of solvents on the basis of various criteria, General properties of ionizing solvents, Different types of chemical reactions taking place in a solvent, Different types of chemical reactions taking place in liquid NH_3 and their comparison to those taking place in aqueous medium , liquid SO_2 as a solvent, liquid HF as a solvent, liquid N_2O_4 as a solvent and glacial acetic acid as a solvent.

References:

1. Purcell, K.F & Kotz, J.C. *Inorganic Chemistry* W.B. Saunders Co, 1977. Huheey, J.E., *Inorganic Chemistry*, Prentice Hall, 1993.
2. Lippard, S.J. & Berg, J.M. *Principles of Bioinorganic Chemistry* Panima Publishing Company 1994.
3. Cotton, F.A. & Wilkinson, G, *Advanced Inorganic Chemistry* Wiley-VCH, 1999
4. Basolo, F, and Pearson, R.C. *Mechanisms of Inorganic Chemistry*, John Wiley & Sons, NY, 1967. Greenwood, N.N. & Earnshaw A. *Chemistry of the Elements*, Butterworth-Heinemann, 1997.

SEMESTER-VI

MAJOR COURSE- MJ-13: BIO-ORGANIC CHEMISTRY: AMINO ACIDS, LIPIDS, ENZYMES, NUCLEIC ACIDS & PHARMACEUTICAL COMPOUNDS

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

On completion of this course, the students will be able to understand

Learning Objectives:

1. Nature of amino acids, peptides and proteins and their biological significance.
2. Enzymes and their importance.
3. Nucleic acids and their biological role.
4. Structure of pharmaceutical compounds and their biological activity.

Course Learning Outcomes:

On successful completion of this course the student should know:

1. Elucidating reaction mechanisms for organic reactions.
2. Nucleic acids and heredity
3. Lipids and their health implications
4. Enzymes and their applications.
5. Pharmaceutical compounds and their applications.

UNIT-1: Amino acids, Peptides and Proteins: (20 Lectures)

Amino acids: Classification of Amino Acids, α -Amino acids- synthesis, ionic properties and reactions, Zwitter-ion structure, pK_a -values, isoelectric point and electrophoresis.

Study of peptides: Determination of primary structure by end group analysis, Synthesis of peptides by using N-protecting, C-protecting and C-activating groups, Merrifield solid phase peptide synthesis.

Proteins: General characteristics, Classification of proteins, Overview of Primary, Secondary, Tertiary and Quaternary structure of proteins.

UNIT-2: Enzymes: (10 Lectures)

General characteristics, Mechanism of enzyme action, factors affecting enzyme action, Coenzymes and cofactors and their role in biological reactions, Specificity of enzyme action (including stereospecificity).

Enzyme inhibitors and their importance, phenomenon of inhibition (competitive and non-competitive inhibition including allosteric inhibition).

UNIT-3: Lipids (10 Lectures)

Introduction to lipids, classification. Oils and fats: Common fatty acids present in oils and fats, Omega fatty acids, Trans fats, Hydrogenation, Saponification value, Iodine number. Biological importance of triglycerides, phospholipids, glycolipids, and steroids (cholesterol).

UNIT-4: Nucleic Acids (10 Lectures)

Definition and general characteristics, Components of nucleic acids, nucleosides and nucleotides, Structure of nucleic acids, The chemical basis of heredity- Replication of DNA, Synthesis of adenine, guanine, cytosine, uracil and thymine.

UNIT-5: Pharmaceutical Compounds-Structure and Importance:(10 Lectures)

Definition, Classification, Structure and Therapeutic use of

Antipyretics: Paracetamol (with synthesis)

Analgesics: Ibuprofen (with synthesis)

Antimalarials: Chloroquine (with synthesis)

Antibiotics: Chloramphenicol (with synthesis)

Reference Books:

1. P Sykes, A Guide Book to Mechanism in Organic Chemistry, 6th Edition (1997), Orient Longman, New Delhi.
2. Morrison, R. T., Boyd, R. N., Bhattejee, S.K., Organic Chemistry, 7th Edn., Pearson.
3. Acheson, R.M. Introduction to the Chemistry of Heterocyclic compounds, John Wiley & Sons(1976).
4. Solomons, T.W., Fryhle Craig, Organic Chemistry, John Wiley & Sons, Inc (2009).
5. McMurry, J.E. Fundamentals of Organic Chemistry, 7th Ed. Cengage Learning India Edition,2013.
6. Kalsi, P. S. Organic reactions and their mechanisms, New Age Science (2010).
7. Clayden, J., Greeves, N., Warren, S., Wothers, P., Organic Chemistry, Oxford University PressInc., New York (2001).
8. Singh, J., Ali, S.M. & Singh, J. Natural Product Chemistry, Prajati Parakashan (2010).
9. Bansal R. K. Heterocyclic Chemistry: Syntheses, Reactions and Mechanisms, New Age, ThirdEdition (1999).
10. J. C. Kotz, P. M. Treichel & J. R. Townsend: General Chemistry, Cengage Lening India Pvt. Ltd., New Delhi (2009).
11. B. H. Mahan: University Chemistry 3rd Ed. Narosa (1998).
12. R. H. Petrucci: General Chemistry 5th Ed. Macmillan Publishing Co.: New York (1985).
13. F.A. Cotton & G. Wilkinson: Basic Inorganic Chemistry, John Wiley.
14. Gary Wulfsberg: Inorganic Chemistry, Viva Books Pvt. Ltd.
15. Morrison, R. T. & Boyd, R. N. Organic Chemistry, Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
16. Finar, I. L. Organic Chemistry (Volume 1), Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
17. Finar, I. L. Organic Chemistry (Volume 2), Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
18. Nelson, D. L. & Cox, M. M. Lehninger's Principles of Biochemistry 7th Ed., W. H. Freeman.
19. Berg, J. M., Tymoczko, J. L. & Stryer, L. Biochemistry 7th Ed., W. H. Freeman.

SEMESTER-VI

MAJOR COURSE- MJ 14: PHASE EQUILIBRIA, CHEMICAL KINETICS & SURFACE CHEMISTRY

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE=75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

After completion the course, the learner shall be able to understand:

Learning objective:

1. Phases, components, Gibbs phase rule, Phase diagrams and applications.
2. Chemical kinetics: type of reactions, determination of rate, theories of reaction rate, steady state approximation.
3. Catalyst – mechanism, acid base catalysis, enzyme catalysis.
4. Adsorption isotherms.
5. Understanding phases, components, Gibb's phase rule and its applications, construction of phase diagram of different systems, the application of phase diagram.
6. Understanding the basics of chemical kinetics: determination of order, molecularity, and understanding theories of reaction rates, determination of rate of opposing/parallel/chain reactions with suitable examples, application of steady state kinetics, Steady-state approximation.
7. Catalyst – mechanism of catalytic action, enzyme catalysis.
8. Langmuir, Freundlich – adsorption isotherms, significance, multilayer adsorption – theory and significance.

Couse Outcomes:

1. Application of phase diagram.
2. Study of reaction kinetics, Fast reactions.
3. Heterogeneous catalysis used in industry and its mechanism of action.
4. Application of adsorption isotherms in metal adsorption, significance.

UNIT-1: Phase Equilibria: (28 Lectures)

Concept of phases, components and degrees of freedom, derivation of Gibbs Phase Rule for nonreactive and reactive systems; Clausius-Clapeyron equation and its applications to solid-liquid, liquid-vapour and solid-vapour equilibria, phase diagram for one component systems, with applications. Phase diagrams for systems of solid-liquid equilibria involving eutectic, congruent and incongruent melting points, solid solutions. Three component systems, water-chloroform- acetic acid system, triangular plots. Binary solutions: Gibbs-Duhem-Margules equation, its derivation and applications to fractional distillation of binary miscible liquids (ideal and nonideal), azeotropes, lever rule, partial miscibility of liquids, CST, miscible pairs, steam distillation. Nernst distribution law: its derivation and applications.

UNIT-2: Chemical Kinetics:(18 Lectures)

Order and molecularity of a reaction, rate laws in terms of the advancement of a reaction, differential and integrated rate laws for first, second and fractional order reactions, pseudo-

unimolecular reactions, determination of the order, kinetics of complex reactions (limited to first order): (i) Opposing reactions (ii) parallel reactions and (iii) consecutive reactions and their differential rate equations (steady-state approximation in reaction mechanisms) (iv) chain reactions. Temperature dependence of reaction rates; Arrhenius equation; activation energy. Collision theory of reaction rates, Lindemann mechanism, qualitative treatment of the theory of absolute reaction rates.

UNIT-3: Catalysis: (8 Lectures)

Types of catalyst, specificity and selectivity, mechanisms of catalysed reactions at solid surfaces; effect of particle size and efficiency of nanoparticles as catalysts. Enzyme catalysis, Michaelis-Menten mechanism, acid-base catalysis.

UNIT-4: Surface Chemistry: (6 Lectures)

Physical adsorption, chemisorption, adsorption isotherms (Freundlich, Temkin, Derivation of Langmuir adsorption isotherms, surface area determination), BET theory of multilayer adsorption (no derivation), Adsorption in solution.

Reference Books:

1. Peter Atkins & Julio De Paula, Physical Chemistry 9th Ed., Oxford University Press (2010).
2. Castellan, G. W. Physical Chemistry, 4th Ed., Narosa (2004).
3. McQuarrie, D. A. & Simon, J. D., Molecular Thermodynamics, Viva Books Pvt. Ltd.: New Delhi (2004).
4. Engel, T. & Reid, P. Physical Chemistry 3rd Ed., Prentice-Hall (2012).
5. Assael, M. J.; Goodwin, A. R. H.; Stamatoudis, M.; Wakeham, W. A. & Will, S.
6. Commonly Asked Questions in Thermodynamics. CRC Press: NY(2011).
7. Zundhal, S.S. Chemistry concepts and applications Cengage India(2011).
8. Ball, D. W. Physical Chemistry Cengage India (2012).
9. Mortimer, R. G. Physical Chemistry 3rd Ed., Elsevier: NOIDA, UP(2009).
10. Levine, I. N. Physical Chemistry 6th Ed., Tata McGraw-Hill (2011).
11. Metz, C. R. Physical Chemistry 2nd Ed., Tata McGraw-Hill (2009).

SEMESTER-VI

MAJOR COURSE- MJ 15: MAJOR PRACTICALS-V

Credit: Theory-04, 120 Hours, Full Marks=100, Pass Marks= 40,

Marks: 25 (Attendance=5 + 20 SIE: 1Hr) + 75 (ESE: 6Hrs) = 100, Pass Marks: Practical(SIE + ESE) = 40

Instructions to Question Setter for

Sessional Internal Practical Examination (SIE):

There will be one Sessional Internal Practical Examination of 2 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 20 marks

Attendance=5 Marks

End Semester Examination (ESE):

There will be one Practical Examination of 6 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

Two Experiments = 65 marks

Practical record notebook= 5 marks

Viva-voce= 5 marks

I. Analysis of Salt Mixture

Qualitative semi micro analysis of mixtures containing 3 anions and 3 cations. Emphasis should be given to the understanding of the chemistry of different reactions. The following radicals are suggested: CO_3^{2-} , NO_2^- , S^{2-} , SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, CH_3COO^- , F^- , Cl^- , Br^- , I^- , NO_3^- , BO_3^{3-} , $\text{C}_2\text{O}_4^{2-}$, PO_4^{3-} , NH_4^+ , K^+ , Pb^{2+} , Cu^{2+} , Cd^{2+} , Bi^{3+} , Sn^{2+} , Sb^{3+} , Fe^{3+} , Al^{3+} , Cr^{3+} , Zn^{2+} , Mn^{2+} , Co^{2+} , Ni^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+} , Mg^{2+} . Mixtures should preferably contain one interfering anion, or insoluble component (BaSO_4 , SrSO_4 , PbSO_4 , CaF_2 or Al_2O_3) or combination of anions e.g. CO_3^{2-} and SO_3^{2-} , NO_2^- and NO_3^- , Cl^- and Br^- , Cl^- and I^- , Br^- and I^- , NO_3^- and Br^- , NO_3^- and I^- . (Spot tests should be done whenever possible.)

II. Phase Equilibria and Chemical Kinetic Measurements

1. Distribution of acetic acid / benzoic acid between water & cyclohexane.
2. Determination of critical solution temperature and composition of the phenol-water system and to study the effect of impurities on it.
3. Study of equilibrium of at least one of the following reactions by the distribution method:
a) $\text{I}_2(\text{aq}) + \text{I}^- \rightarrow \text{I}_3^-(\text{aq})$ b) $\text{Cu}^{++}(\text{aq}) + n \text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_n]^{2+}$
4. Study of Kinetics of reaction of:
a) Acid hydrolysis of methyl acetate with HCl. b) Saponification of ethyl acetate.

III. Organic Chemistry

1. Estimation of glycine by Sorenson's formalin method.
2. Study of the titration curve of glycine.
3. Saponification value of an oil or a fat.
4. Determination of Iodine number of an oil/ fat.

Reference Books:

1. Vogel, A.I. A text book of Quantitative Analysis, ELBS 1986.
2. Mann, F.G. & Saunders, B.C. Practical Organic Chemistry, Pearson Education (2009)
3. Furniss, B.S.; Hannaford, A.J.; Smith, P.W.G.; Tatchell, A.R. Practical Organic Chemistry, 5th Ed., Pearson (2012)
4. Ahluwalia, V.K. & Aggarwal, R. Comprehensive Practical Organic Chemistry: Preparation and Quantitative Analysis, University Press (2000).
5. Ahluwalia, V.K. & Dhingra, S. Comprehensive Practical Organic Chemistry: Qualitative Analysis, University Press (2000).
6. Khosla, B. D.; Garg, V. C. & Gulati, A. Senior Practical Physical Chemistry, R. Chand & Co.: New Delhi (2011)

SEMESTER-VII
MAJOR COURSE

MAJOR COURSE- MJ 16: ORGANIC SPECTROSCOPY, CARBOHYDRATES & DYES

Credit: Theory-04, 60 Hours, Full Marks=100, Pass Marks= 40

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

On completion of this course, the students will be able to understand

1. Understanding structure of molecules on the basis of spectroscopic techniques.
2. Understanding the significance and structure of carbohydrates.
3. Understanding the structure, mechanism of action of selected dyes.

Course Learning Outcomes:

On successful completion of this course the student should know:

1. Use of spectroscopic techniques in the elucidation of structure of
2. Structures of monosaccharides, disaccharides and polysaccharides.
3. Synthesis of dyes ,food colours and their applications

UNIT-1: Organic Spectroscopy:(30 Lectures)

General principles and introduction to absorption and emission spectroscopy.

a) UV-VIS Spectroscopy: Basic principle, Absorption laws, Types of electronic transitions, λ_{max} , Chromophores and Auxochromes, Bathochromic and Hypsochromic shifts, Intensity of absorption bands- Hyperchromic and Hypochromic shifts, Effect of solvent on electronic spectra,

Woodward-Fieser rules and their applications in the calculation of λ_{max} of

- a) α , β -unsaturated aldehydes, ketones, carboxylic acids and esters
- b) Conjugated dienes with or without extended conjugation.
- c) Distinction between cis- and trans- isomers

b)IR Spectroscopy: Basic principle, Molecular vibrations, Fingerprint region and its significance, Group frequency region and its importance, Infrared absorption frequencies of organic molecules with functional groups containing O, N and S (such as ketones, aldehydes, esters, amides, acids, anhydrides, lactones, lactams, thiols, thioethers, etc.); Effect of H-bonding, resonance/conjugation and ring-size on IR absorptions.

c)NMR Spectroscopy: Basic principle, Reference in proton magnetic resonance spectroscopy and Chemical shift, Chemical shifts of H atoms bonded to carbon atoms (aliphatic, olefinic aldehydic and aromatic) and other nuclei (alcohols, phenols, enols, carboxylic acids, amines, amides, mercapto), Spin-spin coupling, Chemical exchange, effect of deuteration.

d)Mass Spectrometry: Basics of fragmentations in organic compounds. Discussion of molecular ion peak, base peak and metastable ions, McLafferty rearrangement. Nitrogen rule, Index of hydrogen deficiency. Application of fragmentation in characterization of organic compounds.

e) Electronic Spectroscopy:

Franck-Condon principle, electronic transitions, singlet and triplet states, fluorescence and phosphorescence, dissociation and predissociation.

f) Atomic Absorption spectroscopy: Theory and application (with some example).

g) Problem Solving: Problems on structure elucidation of organic compounds based on spectral data. Applications of IR, UV, NMR and Mass spectra for identification of simple organic molecules. Solving problems for the elucidation of molecular structure with the help of mixed spectral data.

UNIT-2: Carbohydrate: (16 Lectures)

Occurrence, Definition, classification and their biological importance.

a) Monosaccharides: Constitution and absolute configuration of glucose and fructose, epimers and anomers, mutarotation, determination of ring size of glucose and fructose, Haworth projections and conformational structures, Interconversions of aldoses and ketoses, Killiani-Fischer synthesis and Ruff degradation.

b) Disaccharides: Structure elucidation of maltose, lactose and sucrose.

c) Polysaccharides: Elementary treatment of starch, cellulose and glycogen excluding their structure elucidation.

UNIT-3: Dyes, Food Colours: (14 Lectures)

Classification, Colour and constitution, Mordant and Vat Dyes, Chemistry of dyeing, Synthesis and applications of

a) Azo dyes: Methyl Orange and Congo Red (mechanism of Diazo Coupling)

b) Triphenyl Methane Dyes: Malachite Green, Rosaniline and Crystal Violet

c) Phthalein Dyes: Phenolphthalein and Fluorescein,

d) Natural dyes –Structure elucidation and synthesis of Alizarin and Indigotin, Natural and synthetic food colours with examples, Natural food colours carotenoids, chlorophyll, anthocyanin, betanin and turmeric, Preparation of Natural food colours (Red Pink, Green, Orange, yellow, Blue, Purple, Brown, Tan and Black). . Synthesis and study of Tartrazine, Quinoline yellow, Sunset yellow, Amaranth, Erythrosine, Indigo carmine.

Reference Books:

1. Morrison, R. T., Boyd, R. N., Bhatnagar, S.K., Organic Chemistry, 7th Edn., Pearson.
2. Acheson, R.M. Introduction to the Chemistry of Heterocyclic compounds, John Wiley & Sons (1976).
3. Solomons, T.W., Fryhle Craig, Organic Chemistry, John Wiley & Sons, Inc (2009).
4. McMurry, J.E. Fundamentals of Organic Chemistry, 7th Ed. Cengage Learning India Edition, 2013.
5. Clayden, J., Greeves, N., Warren, S., Wothers, P., Organic Chemistry, Oxford University Press Inc., New York (2001).
6. Singh, J., Ali, S.M. & Singh, J. Natural Product Chemistry, Prajati Parakashan (2010).
7. J. C. Kotz, P. M. Treichel & J. R. Townsend: General Chemistry, Cengage Learning India Pvt. Ltd., New Delhi (2009).
8. B. H. Mahan: University Chemistry 3rd Ed. Narosa (1998).
9. R. H. Petrucci: General Chemistry 5th Ed. Macmillan Publishing Co.: New York (1985).
10. F.A. Cotton & G. Wilkinson: Basic Inorganic Chemistry, John Wiley.
11. Gary Wulfsberg: Inorganic Chemistry, Viva Books Pvt. Ltd.
12. Morrison, R. T. & Boyd, R. N. Organic Chemistry, Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
13. Finar, I. L. Organic Chemistry (Volume 1), Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
14. Finar, I. L. Organic Chemistry (Volume 2), Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
15. Nelson, D. L. & Cox, M. M. Lehninger's Principles of Biochemistry 7th Ed., W. H. Freeman.
16. Berg, J. M., Tymoczko, J. L. & Stryer, L. Biochemistry 7th Ed., W. H. Freeman
17. Ali, Hashmat, Reaction Mechanism in Organic Chemistry, S Chand

SEMESTER-VII

MAJOR COURSE- MJ 17: ELECTROCHEMISTRY, ELECTRICAL AND MAGNETIC PROPERTIES OF MATERIALS & POLYMERS

Credit: Theory-04, 60 Hours, Full Marks=100, Pass Marks= 40

Marks: 25 (5 Attd. + 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100

Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

After completion of the course, the learner can be able to understand:

On completion of this course, the students will be able to understand:

1. Basic principle of laws of electrochemistry.
2. Understanding about chemical cells and their function.
3. Understanding about electrodes, EMF measurement.
4. Understanding about potentiometric titrations and their applications.
5. Basic ideas about polymers.

Course Learning Outcomes:

On successful completion of this course the student should know:

1. Electrolyte and non-electrolyte.
2. Passage of electric current through the aqueous solution or molten state of electrolyte.
3. Non-spontaneous chemical reactions taking place at electrodes during electrolysis.
4. Mechanism of production of electric current due to spontaneous chemical reactions.
5. Classification and applications of polymers.

UNIT 1: Conductance: (16 Lectures)

Arrhenius theory of electrolytic dissociation. Conductivity, equivalent and molar conductivity and their variation with dilution for weak and strong electrolytes. Molar conductivity at infinite dilution. Kohlrausch law of independent migration of ions. Debye-Huckel-Onsager equation, Wien effect, Debye-Falkenhagen effect, Walden's rules. Ionic velocities, mobilities and their determinations, transference numbers and their relation to ionic mobilities, determination of transference numbers using Hittorf and Moving Boundary methods. Applications of conductance measurement: (i) degree of dissociation of weak electrolytes, (ii) ionic product of water (iii) solubility and solubility product of sparingly soluble salts (iv) hydrolysis constants of salts etc.

UNIT 2: Electrochemistry: (12 Lectures)

Quantitative aspects of Faraday's law. Applications of electrolysis in metallurgy and industry. Half-cell potential, Chemical cells, reversible and irreversible cells with examples. Electromotive force of a cell and its measurement, Nernst equation, Standard electrode (reduction) potential and its application of different kind of half-cells. Electrified interfaces, overpotential, Electrocatalysis- influence of various parameters. Hydrogen electrode.

UNIT 3: Application of EMF measurements: (12 Lectures)

Application of EMF measurements in determining (i) free energy, enthalpy and entropy of a cell reaction, (ii) equilibrium constants, and (iii) pH values, using hydrogen, quinone-hydroquinone, glass and $\text{SbO/Sb}_2\text{O}_3$ electrodes. Concentration cells with and without transference, liquid

junction potential, determination of activity coefficients and transference numbers. Qualitative discussion of potentiometric titrations (acid-base, redox, precipitation).

UNIT 4: Electrical & Magnetic Properties of Atoms and Molecules: (8 Lectures)

Basic ideas of electrostatics, Electrostatics of dielectric media, Clausius-Mosotti equation, Lorenz-Laurentz equation, Dipole moment and molecular polarizabilities and their measurements. Diamagnetism, paramagnetism, magnetic susceptibility and its measurement, molecular interpretation.

UNIT 5: Polymers:(12 Lectures)

a)Introduction: Polymers, Monomers, Polymerization processes (addition polymerisation & Condensation polymerisation) and their mechanism, Classification of polymers (elastomers, thermoplastics, thermosetting plastics & fibres; natural, synthetic & semi-synthetic; organic & inorganic; homo polymers & copolymers, isotactic, atactic & syndiotactic), Structure of polymers (linear, branched, cross-linked & network polymers).

b) Properties of polymers: Polydispersity index, crystallinity in polymer, Density, Melting temperature & glass transition temperature, Co-efficient of linear thermal expansion & volumetric thermal expansion.

c)Molecular weight of polymers: Number average molecular weight, Weight average molecular weight, Determination of molecular weight of polymers by light scattering, osmometry, end-group analysis & viscosity measurements

Reference Books:

1. Atkins, P.W & Paula, J.D. Physical Chemistry, 10th Ed., Oxford University Press (2014).
2. Castellan, G. W. Physical Chemistry 4th Ed., Narosa (2004).
3. Mortimer, R. G. Physical Chemistry 3rd Ed., Elsevier: NOIDA, UP (2009).
4. Barrow, G. M., Physical Chemistry 5th Ed., Tata McGraw Hill: New Delhi (2006).
5. Engel, T. & Reid, P. Physical Chemistry 3rd Ed., Prentice-Hall (2012).
6. Rogers, D. W. Concise Physical Chemistry Wiley (2010).
7. Silbey, R. J., Alberty, R. A. & Bawendi, M. G. Physical Chemistry 4th Ed., John Wiley & Sons, Inc. (2005).
8. D.W. Van Krevelen and P.J. Hoftyzen, Properties of Polymer, 3rd Edition Elsevier Scientific, Publishing Company Amsterdam - Oxford - Newyork. 1990.
9. J.E. Mark Ed.AIP, Physical Properties of Polymers Hand Book, Williston, Vt, 1996.
10. S K Gupta and Anil Kumar, Reaction Engineering of Step Growth Polymerization, Plenum Press, 1987
11. Odian, George, Principles of Polymerization, McGraw-Hill Book Co., New York (1970).
12. W. Billmeyer, Text book of polymer science, 3rd Edn., 2007, Wiley.
13. J.R.Fried, Polymer Science and Technology, (2005), PHI publication.
14. Billmeyer Jr., Fred W., Textbook of Polymer Science, Wiley- Interscience Publishers, New York (1962).
15. R. S. Drago, 1992, Physical methods for chemistry: Saunders college publication.
16. P. J. Flory, Principle of polymer chemistry, Cornell University Press.
17. P. Ghosh, Polymer Science and technology, Plastics, Rubber and composites, Tata McGraw Hill.
18. V. Gowriker, N. V. Viswanathan, J. Sreedhar, Polymer Science, New Age Int. Publication, 2019.

SEMESTER-VII

Major Course

MAJOR COURSE- MJ 18: QUANTUM CHEMISTRY, MOLECULAR SPECTROSCOPY & PHOTOCHEMISTRY

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

On completion of this course, the students will be able to understand:

1. The wave-particle duality.
2. Schrodinger's wave equation and its application to hydrogen atom.
3. Rotational spectra and its applications.
4. Vibrational spectra and its applications.
5. Raman spectra and their superiority over IR spectra.
6. Electronic spectra and effect of rotations and vibrations.
7. Photochemical reactions and their kinetics.

Course Learning Outcomes:

On successful completion of this course the student should know:

1. Wave like character of extremely small fast-moving particles.
2. Postulates of quantum mechanics and their applications in dealing with small moving particles.
3. Rotational, vibrational, Raman and electronic spectra and their applications in determining molecular parameters.
4. Photochemical reactions and their quantum yields.

UNIT-1: Introduction to Quantum Chemistry: (8 Lectures)

Introduction to black-body radiation and distribution of energy, photo-electric effect, concept of quantization, wave particle duality (de-Broglie's hypothesis), Plank's Quantum theory. The uncertainty principle, the wavefunction: wave function and its interpretation, conditions of normalization and Orthogonality and its significance. Basic idea about operators, eigen function and eigen values.

UNIT 2: The Schrodinger Wave Equation (12 Lectures)

Postulates of quantum mechanics, the Schrodinger wave equation. Discussion of solutions of the Schrodinger equation to some model systems viz., particle in one dimensional box, three-dimensional box, the harmonic oscillator, the rigid rotor and the hydrogen atom. Schrodinger equation in spherical polar coordinates and separation of $R(r)$, $\Theta(\theta)$ & $\Phi(\phi)$ (radial and angular parts), degeneracies, spherical harmonics of the hydrogen atoms.

UNIT-3: Molecular Spectroscopy:(30 Lectures)

Quantization of molecular energies, Boltzmann distribution, Interaction of electromagnetic radiation with molecules and various types of spectra, Born- Oppenheimer approximation.

a)Rotation spectroscopy: The rotation of molecules and classification of molecules on the basis of principal moments of inertia, Rotational spectra of rigid diatomic molecules, Selection rules, Intensities of spectral lines, Determination of bond lengths of diatomic and linear triatomic molecules, Effect of isotopic substitution on rotational spectra, Centrifugal distortion and rotational spectra of non-rigid rotor.

b)Vibrational spectroscopy: Classical equation of vibration, computation of force constant, amplitude of diatomic molecular vibrations, anharmonicity, Morse potential, dissociation energies, fundamental frequencies, overtones, hot bands, degrees of freedom for polyatomic molecules, modes of vibration, concept of group frequencies, Breakdown of Born-Oppenheimer approximation – interaction between rotations and vibrations of molecules, Vibration-rotation spectroscopy, diatomic vibrating rotator, P, Q, R branches.

c)Raman spectroscopy: Introduction, Raman effect, Classical and quantum theories of Raman effect, Rotational Raman spectra, Effect of nuclear spin on rotational Raman spectra, Vibrational Raman spectra, Stokes & anti-Stokes lines and their intensity difference, Rule of mutual exclusion.

d)Electronic Spectroscopy: Franck-Condon principle, Electronic transitions, Selection rules for electronic transitions, Electronic spectra of diatomic molecules, Vibrational coarse structure of electronic spectra, Rotational coarse structure of electronic spectra, Singlet & triplet states of molecules and their characteristics, Fluorescence and phosphorescence, Dissociation and predissociation, Electronic spectra of polyatomic molecules.

UNIT-4: Photochemistry:(10 Lectures)

Laws of photochemistry, Quantum yield, Jablonski diagrams, Franck-Condon principle, Law of photochemical equivalence, Quantum efficiency, Low and high quantum efficiency, Kinetics of photochemical reactions ($H_2 + Br_2 = HBr$, $2HI = H_2 + I_2$), energy transfer in photochemical reactions (photosensitization and quenching), fluorescence, phosphorescence, chemiluminescence, Discussion of Electronic spectra and photochemistry (Lambert-Beer law and its applications).

Reference Books:

1. Laideler K. J. and Meiser J. M. Physical Chemistry Third Edition (International)1999.
2. Levine I. N., Physical Chemistry, Fourth Edition), McGraw-Hill (International), 1995.
3. McQuarrie D. A. and Simon J. D. Physical Chemistry- A Molecular Approach, University Science Books, 1998.
4. Chandra, A. K. Introductory Quantum Chemistry Tata McGraw-Hill (2001).
5. House, J. E. Fundamentals of Quantum Chemistry 2nd Ed. Elsevier: USA (2004).
6. Laideler K. J. and Meiser J. M. Physical Chemistry Third Edition (International)1999.
7. Rohatgi-Mukherjee K. K. Fundamentals of Photochemistry, New age (revised second edition).
8. Banwell, C. N. & McCash, E. M. Fundamentals of Molecular Spectroscopy 4th Ed. Tata McGraw-Hill: New Delhi (2006).

SEMESTER-VII

Major Course

MAJOR COURSE- MJ 19: MAJOR PRACTICALS-VI

Credit: Theory-04, 120 Hours, Full Marks=100, Pass Marks= 40,

Marks: 25 (Attendance=5 + 20 SIE: 1Hr) + 75 (ESE: 6Hrs) = 100, Pass Marks: Practical (SIE + ESE) = 40

Instructions to Question Setter for

Sessional Internal Practical Examination (SIE):

There will be one Sessional Internal Practical Examination of 2 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 20 marks

Attendance=5 Marks

End Semester Examination (ESE):

There will be one Practical Examination of 6 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

Two Experiments = 65 marks

Practical record notebook= 5 marks

Viva-voce= 5 marks

I. Conductometry

- i. Determination of cell constant
- ii. Determination of equivalent conductance, degree of dissociation and dissociation constant of a weak acid.
- iii. Perform the following conductometric titrations:
 1. Strong acid vs. strong base
 2. Weak acid vs. strong base
 3. Mixture of strong acid and weak acid vs. strong base
 4. Strong acid vs. weak base
 5. Construction of Daniell cell and measurement of EMF.

II. Polymer synthesis

1. Preparation of nylon 66/6
2. Preparation of urea-formaldehyde resin.

III. Estimations

1. Determination of temporary hardness in supplied sample of water.
2. Determination of permanent hardness in supplied sample of water.
3. Determination of total hardness of water by Complexometry.
4. Estimation of Magnesium and Calcium in a mixture by Complexometry.
5. Estimation of Copper & Zn in mixture by Gravimetry.
6. Estimation of Cu & Ni in a mixture by Gravimetry.

IV. Preparations and Characterization

1. Grignard preparation of dye (malachite green (using methylbenzoate)/crystal violet (using diethylcarbonate) (starting material as p-bromo N, N-dimethyl aniline) (only demonstration purpose)
2. Preparation of various Schiff base-metal complexes and their identification using spectroscopy.
3. Preparation of any two of the following complexes and measurement of their conductivity measurement:
 - a. tetraamminecarbonatocobalt (III) nitrate
 - b. tetraamminecopper (II) sulphate
 - c. potassium trioxalatoferrate (III) trihydrate

Reference Books:

1. Vogel, Arthur I: A Test book of Quantitative Inorganic Analysis (Rev. by G.H Jeffery and others) 5th Ed. The English Language Book Society of Longman.

2. Willard, Hobert H. et al.: Instrumental Methods of Analysis, 7th Ed. Wardsworth Publishing Company, Belmont, California, USA, 1988.
3. Ditts, R.V. Analytical Chemistry – Methods of separation
4. Malcolm P. Stevens, Polymer Chemistry: An Introduction, 3rd Ed.
5. Harry R. Allcock, Frederick W. Lampe and James E. Mark, Contemporary Polymer Chemistry, 3rd ed. PrenticeHall (2003)
6. Fred W. Billmeyer, Textbook of Polymer Science, 3rd ed. Wiley-Interscience (1984)
7. Joel R. Fried, Polymer Science and Technology, 2nd ed. Prentice-Hall (2003)
8. Petr Munk and Tejraj M. Aminabhavi, Introduction to Macromolecular Science, 2nd ed. John Wiley & Sons (2002)
9. L. H. Sperling, Introduction to Physical Polymer Science, 4th ed. John Wiley & Sons (2005)
10. Malcolm P. Stevens, Polymer Chemistry: An Introduction, 3rd ed. Oxford University Press (2005)
10. Seymour/ Carraher's Polymer Chemistry, 9th ed. by Charles E. Carraher, Jr. (2013).
11. Synthesis of organometallic compounds: A practical guide, S. Komiya, Wiley.

SEMESTER-VIII
Major Course

MAJOR COURSE- MJ 20: ORGANOMETALLIC AND BIOINORGANIC CHEMISTRY, OR

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

On completion of this course, the students will be able to understand:

1. 18-electron rule and its applications.
2. Organometallic compounds and their applications in organic synthesis and industries.
3. Bioinorganic chemistry – metal ions in biological system, its toxicity.
4. Understanding the bioinorganic chemistry of metals in biological systems.
5. Hemoglobin and its importance in biological systems

Course Learning Outcomes:

On successful completion of this course the student should know:

1. Action of organometallic compounds in accelerating rate of chemical reactions.
2. Role of Mg^{2+} during photosynthesis by green plants.
3. Toxicity of various metals and mechanism of metal-biological system interaction.
4. Transport and storage of O_2 in human beings.

UNIT 1: Organometallic Compounds: (10 Lectures)

Definition and classification of organometallic compounds on the basis of bond type. Concept of hapticity of organic ligands. Metal carbonyls: 18 electron rule, electron count of mononuclear, polynuclear and substituted metal carbonyls of 3d series. General methods of preparation (direct combination, reductive carbonylation, thermal and photochemical decomposition) of mono and binuclear carbonyls of 3d series.

UNIT 2: Synergic effects: (14 Lectures)

EAN rule as applied to carbonyls. Preparation, structure, bonding and properties of mononuclear and polynuclear carbonyls of 3d metals. p-acceptor behaviour of carbon monoxide. Synergic effects (VB approach)- (MO diagram of CO can be referred to for synergic effect to IR frequencies). Structures of mononuclear and binuclear carbonyls of Cr, Mn, Fe, Co and Ni using VBT. pi-acceptor behaviour of CO (MO diagram of CO to be discussed), synergic effect and use of IR data to explain extent of back bonding. Definition and Classification with appropriate examples based on nature of metal-carbon bond (ionic, s, p and multicentre bonds). Structures, properties and reactions of organometallic compounds of Mg, Al, Sn and Li –Use in synthesis of organic compounds.

UNIT-3: Ferrocene & Zeise's salt: (10 Lectures)

Preparation and reactions (acetylation, alkylation, metallation, Mannich Condensation). Structure and aromaticity. Comparison of aromaticity and reactivity with that of benzene. Preparation & structure of Zeise's salt. Evidences of synergic effect and comparison of synergic effect with that in carbonyls.

UNIT-4: Metal Alkyls: (6 Lectures)

Important structural features of methyl lithium (tetramer) and trialkyl aluminium (dimer), concept of multicentre bonding in these compounds. Role of triethylaluminium in polymerisation of ethene (Ziegler – Natta Catalyst). Species present in ether solution of Grignard reagent and their structures, Schlenk equilibrium.

UNIT-5: Bioinorganic Chemistry: (12 Lectures)

A brief introduction to bio-inorganic chemistry. Geochemical effect on distribution of metals. Role of metal ions present in biological systems with special reference to Na^+ , K^+ and Mg^{2+} ions: Na/K pump, Role of Mg^{2+} ions in energy production and chlorophyll. Iron and its application in bio- systems, Haemoglobin, Myoglobin, Storage and transfer of iron. Role of Ca^{2+} in blood clotting, stabilization of protein structures and structural role (bones).

UNIT 6: Catalysis by Organometallic Compounds (8 Lectures)

Study of the following industrial processes and their mechanism:

1. Alkene hydrogenation (Wilkinson's Catalyst)
2. Hydroformylation (Co salts)
3. Wacker Process
4. Synthetic gasoline (Fischer Tropsch reaction)
5. Synthesis gas by metal carbonyl complexes

Reference Books:

1. Lippard, S.J. & Berg, J.M. Principles of Bioinorganic Chemistry Panima Publishing Company 1994.
2. Cotton, F.A. & Wilkinson, G, Advanced Inorganic Chemistry Wiley-VCH, 1999
3. Basolo, F, and Pearson, R.C. Mechanisms of Inorganic Chemistry, John Wiley & Sons, NY, 1967.
4. Greenwood, N.N. & Earnshaw A. Chemistry of the Elements, Butterworth-Heinemann, 1997
5. Powell, P. Principles of Organometallic Chemistry, Chapman and Hall, 1988.
6. Shriver, D.D. & P. Atkins, Inorganic Chemistry 2nd Ed., Oxford University Press, 1994.
7. Purcell, K.F. & Kotz, J.C., Inorganic Chemistry, W.B. Saunders Co. 1977
8. Miessler, G. L. & Donald, A. Tarr, Inorganic Chemistry 4th Ed., Pearson, 2010.
9. Collman, James P. et al. Principles and Applications of Organotransition Metal Chemistry. Mill Valley, CA: University Science Books, 1987.
10. Crabtree, Robert H. The Organometallic Chemistry of the Transition Metals. New York, NY: John Wiley, 2000.
11. Spessard, Gary O., & Gary L. Miessler. Organometallic Chemistry. Upper Saddle River, NJ: Prentice-Hall, 1996.

OR
SEMESTER-VIII

Major Course

MAJOR COURSE- MJ 20: SEMINAR & GROUP DISCUSSION

Credit: Theory-04, Full Marks=100, Pass Marks= 40,

Course Objectives:

1. To enhance students' communication skills through active participation in seminars and group discussions.
2. To prepare students for professional and academic settings where effective communication and collaboration are essential.

Course Outcomes:

Students will be prepared to apply the communication and collaboration skills developed in this course to professional and academic contexts, enhancing their overall employability and academic success.

Course Contents:

The Department Head will organize students into various groups, assigning each group topics from different branches of Chemistry. Under the guidance of a departmental teacher, students will collaborate on their respective topics. Each and every student has to submit the soft copy and the electronically typed hardbound copy of the dissertation with the Plagiarism Report strictly adhering to the University Grants Commission (Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions) Regulations, 2018 F. 1-18/2010(CPP-II). Published in THE GAZETTE OF INDIA: EXTRAORDINARY [PART III—SEC. 4] dated 23rd July, 2018 [web link :https://www.ugc.gov.in/pdfnews/7771545_academic-integrity-Regulation2018.pdf] and the originality declarations of the supervisor and the student along with the raw data at least a week before the examination.

Steps of Evaluation:

Group discussion will be held within each group under the chairmanship of Head of the Department at the department level.

Power point Presentation and the report (as per the standard uniform format provided by the Department) submitted by the students shall be evaluated by one external member and one internal member.

External Member will be appointed by the Head of Department from the following:

Permanent professors working in the postgraduate department of the university or other colleges.

OR

Retired Professor/Associate Professor/Assistant Professor of the university.

Evaluation Process:

Evaluation of the dissertation work may be as per the following guidelines:

Group Discussion	=25 marks
Dissertation Report	= 25 marks
Presentation	= 50 marks

SEMESTER-VIII

ADVANCED MAJOR COURSE- AMJ 1: ANALYTICAL CHEMISTRY

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

After completion of the course, the student shall be able to understand:

Learning objective:

1. Familiarization with fundamentals of analytical chemistry.
2. Basics of spectroscopic, thermal, electrochemical techniques
3. Learning basics of separation techniques and its applications.
4. Understanding analytical tools, statistical methods applied to analytical chemistry.
5. Understanding principle of UV-Vis spectroscopy and its applications.
6. Understanding principles of thermo-gravimetric analysis and study of thermal decomposition of materials/characterization of materials.
7. Understanding basics of electro-analytical techniques and its applications.
8. Understanding principles of separation technology and its use in advanced instrumentations.

Self-study:

1. Thermo-gravimetric Analysis of different compounds and application of mathematical models.
2. Study of different kinds of chromatograms; calculation of R_f ,
3. Analysis of GC/HPLC data for known materials/compounds.

Unit-1: Qualitative and Quantitative Aspects of Analysis: (4 Lectures)

Tools in analytical chemistry and their applications, Sampling, evaluation of analytical data, errors, accuracy and precision, statistical test of data; F, Q and t-test, rejection of data, and confidence intervals.

UNIT-2: Spectroscopy Instrumentation (10 Lectures)

Origin of spectra, interaction of radiation with matter, fundamental laws of spectroscopy and selection rules, validity of Beer-Lambert's law.

Vibration spectroscopy: Basic principles of instrumentation, sampling techniques. Application of IR spectroscopy for characterization through interpretation of data, Effect and importance of isotope substitution. Introduction to Raman spectra

UV-Visible Spectrometry: Basic principles of instrumentation, principles of quantitative analysis using estimation of metal ions from aqueous solution, Determination of composition of metal complexes using Job's method of continuous variation and mole ratio method.

UNIT-3: Thermal Analysis (4 Lectures)

Theory of thermogravimetry (TG and DTG), instrumentation, estimation of Ca and Mg from their mixture.

UNIT-4: Electroanalytical Methods (5 Lectures)

Classification of electroanalytical methods, basic principle of pH metric, potentiometric and

conductometric titrations. Techniques used for the determination of equivalence points.
determination of pK_a values.

UNIT-5: Separation Techniques (17 Lectures)

Solvent extraction: Classification, principle and efficiency of the technique. Mechanism of extraction: extraction by solvation and chelation. Technique of extraction: batch, continuous and counter current extractions. Qualitative and quantitative aspects of solvent extraction: extraction of metal ions from aqueous solution, extraction of organic species from the aqueous and non-aqueous media.

Chromatography techniques: Classification, principle and efficiency of the technique. Mechanism of separation: adsorption, partition & ion exchange. Development of chromatograms: frontal, elution and displacement methods. Qualitative and quantitative aspects of chromatographic methods of analysis using LC, GLC, TLC and HPLC.

UNIT 6: Sample Analysis: (20 Lectures)

Analysis of soil: Composition of soil, Concept of pH and pH measurement, Complexometric titrations, Chelation, Chelating agents, use of indicators

- a. Determination of pH of soil samples.
- b. Estimation of Calcium and Magnesium ions as Calcium carbonate by complexometric titration.

Analysis of water: Definition of pure water, sources responsible for contaminating water, water sampling methods, water purification methods.

- a. Determination of pH, acidity and alkalinity of a water sample.
- b. Determination of dissolved oxygen (DO) of a water sample.

Analysis of food products: Nutritional value of foods, idea about food processing and food preservations and adulteration.

- a. Identification of adulterants in some common food items like coffee powder, asafoetida, chilli powder, turmeric powder, coriander powder and pulses, etc.
- b. Analysis of preservatives and colouring matter.

Analysis of cosmetics: Major and minor constituents and their function

- a. Analysis of deodorants and antiperspirants, Al, Zn, boric acid, chloride, sulphate.
- b. Determination of constituents of talcum powder: Magnesium oxide, Calcium oxide, Zinc oxide and Calcium carbonate by complexometric titration.

Reference Books:

Recommended Books/Reference Books:

- 1 Mendham, J., A. I. Vogel's Quantitative Chemical Analysis 6th Ed., Pearson, 2009.
- 2 Willard, H.H. et al.: Instrumental Methods of Analysis, 7th Ed. Wardsworth Publishing California, USA, 1988.
- Christian, G.D, Analytical Chemistry, 6th Ed. John Wiley & Sons, New York, 2004.
- 4 Harris, D.C.: Exploring Chemical Analysis, 9th Ed. New York, W.H. Freeman, 2016.
- 5 Skoog, D.A. Holler F.J. & Nieman, T.A. Principles of Instrumental Analysis, Saunder College Publications, (1998).
- 6 Mikes, O. Laboratory Hand Book of Chromatographic & Allied Methods, Elles Harwood John Wiley 1979.
- 7 Ditts, R.V. Analytical Chemistry; Methods of separation, van Nostrand, 1974.
- 8 Khopkar, S. M., Basic Concepts of Analytical Chemistry, New Age (Second edition) 1998
9. Skoog D.A., Holler F.J., Nieman T.A., Principles of instrumental analysis, 5th Edn., Brooks & Cole (1997).

SEMESTER-VIII

ADVANCED MAJOR COURSE- AMJ 2: GREEN CHEMISTRY & NANO CHEMISTRY

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

Learning objective:

1. Green chemistry and its principles.
2. Green synthesis and reactions.
3. Green chemistry for sustainable solutions.
4. Understanding principles of green chemistry.
5. Understanding design of chemical reactions/chemical synthesis using green chemistry principles.
6. Atom economy and design of chemical reactions using the principle.
7. Understanding the use of green chemistry principle and processes in laboratory reactions.
8. Nanoscience, Nanostructure and Nanotechnology
9. Size Dependent Properties of Nanomaterials
10. Synthesis of Nanomaterials

Learning Outcomes:

1. Use of green chemistry in designing new laboratory experiments.
2. Use of principle of atom economy and design experiments using the principle.
3. Use of Nanomaterials

Green Chemistry

UNIT-1: Introduction to Green Chemistry:(3 Lectures)

What is Green Chemistry? Need for Green Chemistry. Goals of Green Chemistry. Limitations/ Obstacles in the pursuit of the goals of Green Chemistry.

UNIT -2: Principles of Green Chemistry and Designing Green Synthesis:(15 Lectures)

Twelve principles of Green Chemistry with their explanations and examples; Designing a Green Synthesis using these principles; Prevention of Waste/ by products; maximum incorporation of the materials used in the process into the final products (Atom Economy); prevention/ minimization of hazardous/ toxic products; designing safer chemicals – different basic approaches to do so; selection of appropriate auxiliary substances (solvents, separation agents), green solvents, solvent less processes, immobilized solvents and ionic liquids.

Energy requirements for reactions - use of microwaves, ultrasonic energy; selection of starting materials; avoidance of unnecessary derivatization – careful use of blocking/protecting groups; use of catalytic reagents(whenever possible) in preference to stoichiometric reagents;

designing of biodegradable products; prevention of chemical accidents; strengthening/development of analytical techniques to prevent and minimize the generation of hazardous substances in chemical processes.

UNIT 3: Examples of Green Synthesis/ Reactions (22 Lectures)

1. Green Synthesis of the following compounds: adipic acid, catechol, BHT, methyl methacrylate, urethane, aromatic amines (4-aminodiphenylamine), benzyl bromide, acetaldehyde, disodium iminodiacetate (alternative to Strecker synthesis), citral, ibuprofen, paracetamol, furfural.

2. Microwave assisted reactions in water: Hofmann Elimination, Hydrolysis (of benzyl chloride, benzamide, n-phenyl benzamide, methyl benzoate to benzoic acid), Oxidation (of toluene, alcohols).

Microwave assisted reactions in organic solvents: Esterification, Fries rearrangement, Orthoester Claisen Rearrangement, Diels-Alder Reaction, Decarboxylation.

Microwave assisted solid state reactions: Deacetylation, Deprotection. Saponification of esters, Alkylation of reactive methylene compounds, reductions, synthesis of nitriles from aldehydes; anhydrides from dicarboxylic acid; pyrimidine and pyridine derivatives; 1,2-dihydrotriazine derivatives; benzimidazoles.

3. Ultrasound assisted reactions: Esterification, saponification, substitution reactions, Alkylation, oxidation, reduction, coupling reaction, Cannizzaro reaction, Strecker synthesis, Reformatsky reaction.

Nano chemistry

UNIT-4: Introduction to Nanoscience, Nanostructure and Nanotechnology:(10 Lectures)

Basic idea; Overview of nanostructures and nano-materials, classification, (cluster, colloid, nanoparticles, and nanostructures, Spheroid, Wire, Rod, Tube, and Quantum Dot. Carbon nanotubes and inorganic nanowires. Calculation of percentage of surface atom and surface to volume ratio of spherical, wire, rod and disc shapes nanoparticles. Examples of use of nanomaterials in environmental remediation, medicines and industry.

UNIT 5: Size Dependent Properties of Nanomaterials: (5 Lectures)

Basic idea with few examples only: Quantum confinement, Electrical, Optical (Surface Plasmon resonance), variation in colours (Blue shift & Red shift), Magnetic, thermal and catalytic properties.

UNIT 6: Synthesis of Nanomaterials: (5 Lectures)

Brief introduction about Top-down and Bottom-up approaches & self-assembly techniques of nanoparticles synthesis, Solvothermal process, Examples of preparation of gold and silver metallic nanoparticles, self-assembled nanostructures-control of nanoarchitecture-one dimensional control. Carbon nanotubes and inorganic nanowires.

Reference Books:

1. V.K. Ahluwalia & M.R. Kidwai: New Trends in Green Chemistry, Anamlaya Publishers (2005).
2. P.T. Anastas & J.K. Warner: Oxford Green Chemistry- Theory and Practical, University Press (1998).
3. A.S. Matlack: Introduction to Green Chemistry, Marcel Dekker (2001).
4. M.C. Cann & M.E. Connely: Real-World cases in Green Chemistry, American Chemical Society, Washington (2000).
5. M.A. Ryan & M. Tinnesand, Introduction to Green Chemistry, American Chemical Society, Washington (2002). Zhen Guo and Li Tan, Fundamentals and Applications of Nanomaterials.2009, Artech House, London Publication.
6. C. N. R. Rao, A. Muller, A. K. Cheetam, The Chemistry of Nanomaterials: Synthesis, Properties and Applications, Willey-VCH Verlag, Germany, 2005.
7. G. Cao, Nanostructures and Nanomaterials: Synthesis, Properties and Applications, Imperial College Press,London, 2004
8. R. W. Kelsall, I. W. Hameley, M. Geoghegan, Nanoscale Science and Technology, John

Wiley & Sons, England, 2005

9. 7. Charles P. Poole and Frank J Owens, Introduction to nano technology, Wiley, interscience, 2003.
10. 8. Pradeep, T., A text of book of nanoscience and nanotechnology, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 2012.

SEMESTER-VIII

ADVANCED MAJOR COURSE- AMJ 3: MAJOR PRACTICALS-VII

Credit: Theory-04, 120 Hours, Full Marks=100, Pass Marks= 40,

Marks: 25 (Attendance=5 + 20 SIE: 1Hr) + 75 (ESE: 6Hrs) = 100, Pass Marks: Practical(SIE + ESE) = 40

Instructions to Question Setter for

Sessional Internal Practical Examination (SIE):

There will be one Sessional Internal Practical Examination of 2 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 20 marks

Attendance=5 Marks

End Semester Examination (ESE):

There will be one Practical Examination of 6 Hrs duration.

Evaluation of Practical Examination may be as per the following guidelines:

Two Experiments = 65 marks

Practical record notebook= 5 marks

Viva-voce= 5 marks

A. Colourimetry

I. Verify Lambert-Beer's law and determine the concentration of $\text{CuSO}_4/\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ in a solution of unknown concentration

II. Determine the concentrations of KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ in a mixture.

B. Chromatography:

(i) Paper chromatographic separation of Fe^{3+} , Al^{3+} , and Cr^{3+} .

(ii) Separation and identification of the monosaccharides present in the given mixture (glucose & fructose) by paper chromatography. Reporting the R_f values.

iii. Separate a mixture of Sudan yellow and Sudan Red by TLC technique and identify them on the basis of their R_f values.

(iv) Chromatographic separation of the active ingredients of plants, flowers and juices by TLC

C. Solvent Extractions:

(i) To separate a mixture of Ni^{2+} & Fe^{2+} by complexation with DMG and extracting the Ni^{2+} -DMG complex in chloroform, and determine its concentration by spectrophotometry.

ii. Determine the pH of the given aerated drinks fruit juices, shampoos and soaps.

iii. Determination of Na, Ca, Li in cola drinks and fruit juices using flame photometric techniques.

D. Analysis of soil:

(i) Determination of pH of soil.

(ii) Total soluble salt

E Ion exchange:

(i) Determination of exchange capacity of cation exchange resins and anion exchange resins.

(ii) Separation of metal ions from their binary mixture.

(iii) Separation of amino acids from organic acids by ion exchange chromatography.

F. Spectrophotometry

(i). Determination of pK_a values of indicator using spectrophotometry.

(ii) Structural characterization of compounds by infrared spectroscopy.

(iii) Determination of dissolved oxygen in water.

(iv) Determination of chemical oxygen demand (COD).

(v) Determination of Biological oxygen demand (BOD).

(vi) Determine the composition of the Ferric-salicylate/ ferric-thiocyanate complex by Job's method.

G. Green Chemistry and Nano Chemistry

1. Preparation of biodiesel from vegetable/ waste cooking oil.

2. Use of molecular model kit to stimulate the reaction to investigate how the atom economy illustrates Green Chemistry.

3. Reactions like addition, elimination, substitution and rearrangement may also be studied for the calculation of atom economy.
4. Benzoin condensation using Thiamine Hydrochloride as a catalyst (instead of cyanide).
5. Extraction of D-limonene from orange peel using liquid CO₂ prepared from dry ice.
6. Mechanochemical solvent free synthesis of azomethines
7. Solvent free, microwave assisted one pot synthesis of phthalocyanine Cu(II) complex.
8. Photoreduction of benzophenone to benzopinacol in presence of sunlight.
9. Synthesis of Zinc Oxide nanoparticles.
10. Synthesis of Iron Oxide nanoparticles.
11. Synthesis of Nickel nanoparticle using urea decomposition method.
12. Synthesis of Silver and Gold nanoparticles

References:

1. Mendham, J., A. I. Vogel's Quantitative Chemical Analysis 6th Ed., Pearson, 2009.
2. Willard, H.H. et al.: Instrumental Methods of Analysis, 7th Ed. Wardsworth Publishing Company, Belmont, California, USA, 1988.
3. Christian, G.D. Analytical Chemistry, 6th Ed. John Wiley & Sons, New York, 2004.
4. Harris, D.C. Exploring Chemical Analysis, 9th Ed. New York, W.H. Freeman, 2016.
5. Khopkar, S.M. Basic Concepts of Analytical Chemistry. New Age International Publisher, 2009.
6. Skoog, D.A. Holler F.J. and Nieman, T.A. Principles of Instrumental Analysis, Cengage Learning India Edition.
7. Mikes, O. & Chalmers, R.A. Laboratory Handbook of Chromatographic & Allied Methods, Elsevier Harwood Ltd. London.
8. Ditts, R.V. Analytical Chemistry: Methods of separation. Van Nostrand, New York.
9. Anastas, P.T & Warner, J.C. Green Chemistry: Theory and Practice, Oxford University Press (1998).
10. Kirchoff, M. & Ryan, M.A. Greener approaches to undergraduate chemistry experiment. American Chemical Society, Washington DC (2002).
11. Ryan, M.A. Introduction to Green Chemistry, Tinnesand; (Ed), American Chemical Society, Washington DC (2002).
12. Sharma, R.K.; Sidhwani, I.T. and Chaudhari, M.K. I.K. Green Chemistry Experiment: A monograph, International Publishing ISBN 978-93-81141-55-7 (2013).
13. Cann, M.C. and Connelly, M. E. Real world cases in Green Chemistry, American Chemical Society (2008).
14. Lancaster, M. Green Chemistry: An Introductory Text RSC Publishing, Second Edition, 2010.
15. Pavia, D. L., Lampman, G.M., Kriz, G.S. & Engel, R.G. Introduction to Organic Laboratory Techniques: A Microscale and Macro Scale Approach, W.B.Saunders, 1995.
16. Pradeep T., A text book of nanoscience and nanotechnology, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 2012 edition.

**For BSc Honours with Research Degree One Requires to Select RC-1 and RC-2
instead of AMJ-1, AMJ-2 & AMJ-3**

SEMESTER-VIII

BSc (Research) Course

RC-I-RESEARCH METHODOLOGY IN CHEMISTRY

Credit: Theory-04, Full Marks=100, Pass Marks= 40, Lectures:60

Marks: 25 (5 Attendance+ 20 SIE: 1Hr) + 75 (ESE: 3Hrs) = 100 Pass Marks: Th (SIE + ESE) = 40

Instruction to Question Setter for

Semester Internal Examination (SIE 20+5=25 marks):

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 20 Mark (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 75 marks):

There will be two group of questions A and B. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type seven questions of fifteen marks each, out of which any four are to be answered.

Note: There may be subdivisions in the questions of group B.

Course Objectives:

1. To elucidate the foundational principles of research, including its meaning, objectives, motivation, and utility.
2. To acquaint students with the concept of theory, empiricism, deductive and inductive theories, and the characteristics of the scientific method.
3. To guide students in problem identification and formulation, emphasizing the necessity of defining research problems and the significance of literature review.
4. To introduce students to research design, including its concept, importance, features, and various types such as exploratory, descriptive, and experimental designs.
5. To differentiate between qualitative and quantitative research, exploring the concepts of measurement, causality, generalization, replication, and methods for merging both approaches.
6. To familiarize students with data collection and analysis techniques, including observation, methods of data collection, hypothesis testing, generalization, and interpretation.
7. To explain the concept of measurement in research, addressing issues such as validity, reliability, and different levels of measurement (nominal, ordinal, interval, ratio).
8. To instruct students on sampling techniques, covering concepts like statistical population, sample, sampling frame, sampling error, sample size, and characteristics of a good sample.
9. To equip students with skills for data analysis, including data preparation, univariate and bivariate analysis, and testing hypotheses of association using techniques like cross tabulations and Chi-square tests.
10. To prepare students for interpreting data, writing research papers, understanding ethical issues related to publishing, avoiding plagiarism, and utilizing various tools and techniques for effective research.

Learning Objectives:

1. Understand the foundational concepts of research, including its purpose, motivation, and significance.
2. Differentiate between various theories and methods used in research, such as deductive and inductive reasoning.
3. Develop the ability to identify and formulate research problems, emphasizing the importance of literature review.
4. Comprehend the concept and significance of research design, including its features and different types.
5. Analyze the differences between qualitative and quantitative research methods and understand how they can be merged.
6. Acquire skills in data collection, analysis, interpretation, and presentation using appropriate statistical techniques.
7. Gain knowledge about measurement in research, including issues related to validity, reliability, and levels of measurement.
8. Familiarize oneself with sampling techniques and understand their importance in research.
9. Develop proficiency in data analysis, including preparing data, conducting univariate and bivariate analysis, and interpreting results.
10. Learn the process of writing research papers, understanding ethical considerations in publishing, and utilizing tools for effective research and reporting.

Unit-I: Foundation of Research: 5 Lectures

Meaning, Objectives, Motivation, Utility. Concept of theory, empiricism, deductive and inductive theory. Characteristics of scientific method - understanding the language of research - Concept, Construct, definition, Variable. Research Process

Unit-II: Problem Identification & Formulation: 7 Lectures

Definition and formulating the research problem, Necessity of defining the problem, Importance of literature review in defining a problem, Research Question - Investigation Question - Measurement Issues - Hypothesis - Qualities of a good hypothesis - Null hypothesis & Alternative Hypothesis. Hypothesis Testing - Logic & importance

Unit-III: Research Design: 7 Lectures

Concept and Importance in Research - Features of a good research design - Exploratory Research Design - Concept, Types and uses, Descriptive Research Design - concept, types and uses. Experimental Design - Concept of Independent & Dependent variables.

Unit-IV: Qualitative and Quantitative Research: Qualitative - 2 Lectures

Quantitative Research - Concept of measurement, causality, generalization, replication. Merging the two approaches.

Unit-V: Data Collection and analysis: Execution of the research - 3 Lectures

Observation and Collection of data - Methods of data collection, hypothesis-testing - Generalization and Interpretation.

Unit-VI: Measurement: 5 Lectures

Concept of measurement - what is measured? Problem in measurement in research - Validity and Reliability. Levels of measurement - Nominal, Ordinal, Interval, Ratio.

Unit-VII: Sampling: 6 Lectures

Concept of Statistical population, Sample, Sampling Frame, Sampling Error, Sample size, Non-Response. Characteristics of a good sample. Probability Sample - Simple Random Sample, Systematic Sample, Stratified Random Sample & Multi-stage sampling. Determining size of the sample - Practical considerations in sampling and sample size.

Unit-VIII: Data Analysis: 4 Lectures

Data Preparation - Univariate analysis (frequency tables, bar charts, pie charts, percentages), Bivariate analysis - Cross tabulations and Chi-square test including testing hypothesis of association.

Unit-IX: Interpretation of Data and Paper Writing: 8 Lectures

Layout of a Research Paper, Journals in chemistry, Impact factor of journals, When and where to publish? Ethical issues related to publishing, Plagiarism and Self-Plagiarism. Use of Encyclopaedias, Research Guides, Handbook etc., Academic databases for concerned discipline. Use of tools / techniques for Research: methods to search required information effectively, Reference Management Software like Zotero/Mendeley, Software for paper formatting like LaTeX/MsOffice, software for detection of Plagiarism.

Unit-X: Reporting and Thesis writing: 7 Lectures

Structure and components of scientific reports - Types of report - Technical reports and thesis - Significance - Different steps in the preparation - Layout, Structure and Language of typical reports - Illustrations and tables - Bibliography, referencing and footnotes - Oral presentation - Planning - Preparation - Practice - Making presentation - Use of visual aids - Importance of effective communication

Unit-XI: Application of results and ethics: 6 Lectures

Environmental impacts - Ethical issues - ethical committees - Commercialization - Copy right - royalty - Intellectual property rights and patent law - Trade related aspects of intellectual property Rights - Reproduction of published material - Plagiarism - citation and acknowledgement - citation and acknowledgement - Reproducibility and accountability.

References:

1. Garg, B. L., Karadia, R., Agarwal, F. and Agarwal, U. K., 2002. An introduction to Research Methodology, RBSA Publishers.
2. Kothati, C.R., 1990. Research Methodology: Methods and Techniques. New Age International. 418p.
3. Sinha, S. C. and Dhiman, A. K., 2002. Research Methodology, Ess Publications.
4. Trochim, W. M. K., 2005. Research Methods: the concise knowledge base, Atomic Dog Publishing. 270p
5. Wadehra, B. L. 2000. Law relating to patents, trademarks, copyright designs and geographical indications. Universal Law Publishing.
6. Anthony, M., Graziano, A. M. and Raulin, M. L., 2009. Research Methods: A Process of Inquiry, Allyn and Bacon.
7. Carlos, C. M., 2000. Intellectual property rights, the WTO and developing countries: the TRIPS agreement and policy options. Zed Books, New York.
8. Coley, S. M. and Scheinberg, C. A., 1990, "Proposal Writing", Sage Publications.
9. Day, R. A., 1992. How to Write and Publish a Scientific Paper, Cambridge University Press.
10. Fink, A., 2009. Conducting Research Literature Reviews: From the Internet to Paper. Sage Publications
11. Leedy, P. D. and Ormrod, J. E., 2004 Practical Research: Planning and Design, Prentice Hall.
12. Satarkar, S. V., 2000. Intellectual property rights and Copy right. Ess Publications.

SEMESTER-VIII

BSc Honours with Research Course

RC-II- RESEARCH PROJECT OR DISSERTATION

Credits: -08, Full Marks=200, Pass Marks= 80,

Course Objectives:

1. To facilitate students in conducting independent research projects in the field of Chemistry, allowing them to apply theoretical knowledge to practical experimentation.
2. To guide students in developing research skills, including problem identification, formulation of hypotheses, experimental design, data collection, analysis, and interpretation.
3. To foster critical thinking and analytical skills among students through the exploration of research methodologies and scientific literature.
4. To enhance students' communication abilities by requiring them to effectively present their research findings through written reports and oral presentations.
5. To cultivate ethical awareness and responsibility in research practices, including proper citation, acknowledgment of sources, and adherence to ethical guidelines and regulations.

Course Outcomes:

1. Students will demonstrate proficiency in designing and conducting independent research projects in Chemistry, from problem identification to data analysis.
2. Students will develop advanced skills in critical thinking, analytical reasoning, and problem-solving through the exploration of complex scientific questions.
3. Students will exhibit competency in utilizing various research methodologies and techniques appropriate for addressing specific research questions in Chemistry.
4. Students will effectively communicate their research findings through well-organized written reports and articulate oral presentations.
5. Students will demonstrate ethical conduct in research, including proper citation practices, avoidance of plagiarism, and adherence to ethical guidelines and regulations governing scientific research in Chemistry.

Course Contents:

Students who achieve a minimum of 75% marks across the first six semesters and express a desire to engage in undergraduate research may opt for a research stream during their fourth year of study. The Head of the Department (HOD) is responsible for assigning a supervisor to these students, chosen from the permanent faculty members holding a PhD degree. Subsequently, students, in consultation with their supervisor, select a research problem relevant and problem solving in nature to the local community/Industry, environment and may be related to innovation/novel method/idea with interdisciplinary approach to submit a synopsis or research proposal to the department for review.

Following this, the HOD convenes a meeting of the Departmental Research Council, where students present their synopsis or research proposal for approval or rejection. Upon approval, students commence their research work under the guidance of their supervisor and in accordance with the approved proposal.

At the conclusion of the semester, Each and every student has to submit the soft copy and the electronically typed hardbound copy of the project thesis with the Plagiarism Report strictly adhering to the University Grants Commission (Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions) Regulations, 2018 F. 1-18/2010(CPP-II). Published in THE GAZETTE OF INDIA: EXTRAORDINARY [PART III—SEC. 4] dated 23rd July, 2018 [web link :https://www.ugc.gov.in/pdfnews/7771545_academic-integrity-

[Regulation2018.pdf](#) and the originality declarations of the supervisor and the student along with the raw data as date notified/affixed by the Head of the Department.

The department then forwards the thesis to an external evaluator, who provides a report on the work. Following receipt of the external report, the department shall arrange an open viva voce examination for the student, with both external and internal members present to assess the student's work.

The Department Research Committee shall be constituted from the following persons:

HOD of the Department–Chairmen

1. Dean, Faculty of Science- External member
2. HOD, University Department- External member
3. At list two permanent faculty member of the department (Nominated by HOD)- Members External members can be any of the following:
4. Permanent professors working in the postgraduate department of the university or other colleges who have the qualification to become PhD supervisors.

OR

Retired Professor/Associate Professor/Assistant Professor of the university who has been supervising PhD scholar/s.

OR

Professor/Associate Professor/Assistant Professor of the outside university who has been supervising PhD scholar/s.

Note- Minimum three external examiner lists will be sent by HOD through concerned College Principal to the Controller of Examination, SKMU, Dumka for final approval. In that list priority will be given as per order mentioned above.

The project thesis shall be evaluated under the heads as per the standard LaTeX/Word format made available by the Head of the Department which may include the following sections:

1. **Title Page:**
 - Title of the thesis
 - Author's name
 - Affiliation
 - Date
2. **Abstract:**
 - A brief summary of the research work, including the objectives, methods, results, and conclusions.
3. **Table of Contents:**
 - Lists all the major sections and subsections of the thesis, along with their respective page numbers.
4. **List of Figures and Tables:**
 - Enumerates all the figures and tables included in the thesis, along with their page numbers.
5. **Introduction:**
 - Provides background information on the research topic.
 - States the research problem and objectives.
 - Reviews relevant literature and previous research in the field.
 - Outlines the structure of the thesis.
6. **Experimental Methods:**
 - Describes the experimental procedures and methodologies used in the research.
 - Includes details on materials, equipment, and techniques employed.
 - Provides sufficient information to allow for replication of the experiments.
7. **Results and Discussion:**
 - Presents the findings of the research.

- Organizes results logically, often using figures, tables, and graphs.
- Analyses and interprets the results in the context of the research objectives.
- Discusses the implications of the findings and their significance in the field.

8. Conclusion:

- Summarizes the key findings of the research.
- Restates the research objectives and addresses whether they were achieved.
- Offers insights into the broader implications of the research.
- Suggests avenues for future research.

9. References:

- Lists all the sources cited in the thesis, following a specific citation style (e.g., APA, MLA, Chicago).
- Includes journal articles, books, conference proceedings, and other relevant literature.

10. Appendices:

- Contains additional supplementary information that is relevant to the thesis but not essential to the main text.
- Includes raw data, detailed experimental procedures, calculations, or any other supporting material.

Guidelines for distribution of marks may be as follows or as appropriate:

Assessment of project synopsis: 75 marks

Assessment of project thesis: 100 marks

Viva-voce: 25 marks

CHEMISTRY MINOR COURSES

SEMESTER-I **MINOR COURSES**

MN-CHE-1A(THEORY)-: INTRODUCTORY CHEMISTRY

Credit: Theory-03, 45 Hours, Full Marks=75, Pass Marks= 30

Marks: 15 (5 Attd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs) =75

Pass Marks: Th (SIE + ESE) = 30

Instruction to Question Setter for

Semester Internal Examination (SIE 10+5=15 marks):

There will be two group of questions.

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 10 Mark. Question No.1 will be very short answer type in Group A consisting of five questions of 1 mark each. Group B will contain descriptive type two questions of five marks each, out of which any one to answer. (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 60 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No.2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type five questions of fifteen marks each, out of which any three are to answer.

Note: There may be subdivisions in the questions of group B.

Section A: Physical Chemistry

UNIT I: Chemical Energetics: (8 classes each of 60 minutes duration)

Review of thermodynamics and the Laws of Thermodynamics.

Important principles and definitions of thermochemistry. Concept of standard state and standard enthalpies of formations. Calculation of bond energy, bond dissociation energy from thermochemical data. Statement of Third Law of thermodynamics and calculation of absolute entropies of substances.

UNIT II: Chemical Kinetics: (7 classes each of 60 minutes duration)

The concept of reaction rates. Effect of temperature, pressure, catalyst and other factors on reaction rates. Order and molecularity of a reaction. Derivation of integrated rate equations for zero, first and second order reactions (both for equal and unequal concentrations of reactants). Half-life of a reaction. General methods for determination of order of a reaction. Concept of activation energy and its calculation from Arrhenius equation.

Section B: Inorganic Chemistry

UNIT III: Atomic Structure: (5 classes each of 60 minutes duration)

What is Quantum mechanics? Time independent Schrodinger equation and meaning of various terms in it. Significance of ψ and ψ^2 , Schrodinger equation for hydrogen atom. Radial and angular parts of the hydrogenic wave functions (atomic orbitals) and their variations for 1s, 2s, 2p, 3s, 3p and 3d orbitals (Only graphical representation). Radial and angular nodes and their significance. Radial distribution functions and the concept of the most probable distance with special reference to 1s and 2s atomic orbitals. Significance of quantum numbers, orbital angular momentum and quantum numbers m_l and m_s . Shapes of s, p and d atomic orbitals, nodal planes. Discovery of spin, spin quantum number (s) and magnetic spin quantum number (m_s).

Rules for filling electrons in various orbitals, Electronic configurations of the atoms. Stability of half-filled and completely filled orbitals, concept of exchange energy. Relative energies of atomic orbitals, Anomalous electronic configurations.

UNIT IV: Chemical Bonding and Molecular Structure: (10 classes each of 60 minutes duration)

Ionic Bonding: General characteristics of ionic bonding. Energy considerations in ionic bonding, lattice energy and solvation energy and their importance in the context of stability and solubility

of ionic compounds. Polarizing power and polarizability. Fajan's rules, ionic character in covalent compounds, bond moment, dipole moment and percentage ionic character.

Covalent bonding: VB Approach: Shapes of some inorganic molecules and ions on the basis of VSEPR and hybridization with suitable examples of linear, trigonal planar, square planar, tetrahedral, trigonal bipyramidal and octahedral arrangements.

Section C: Organic Chemistry

UNIT V: Fundamentals of Organic Chemistry: (3 classes each of 60 minutes duration)

Inductive Effect, Electromeric Effect, Resonance and Hyperconjugation. Cleavage of Bonds: Homolysis and Heterolysis. Structure, shape and reactivity of organic molecules: Nucleophiles and electrophiles. Reactive Intermediates: Carbocations, Carbanions and free radicals.

Aromaticity: Benzenoids and Hückel's rule.

UNIT VI: Aliphatic hydrocarbons:

Alkanes: (4 classes each of 60 minutes duration) (Upto 5 Carbons)

Preparation: Catalytic hydrogenation, Wurtz reaction, Kolbe's synthesis, from Grignard reagent.

Reactions: Free radical Substitution: Halogenation

Alkenes: (3 classes each of 60 minutes duration) (Upto 5 Carbons)

Preparation: Elimination reactions: Dehydration of alkenes and dehydrohalogenation of alkyl halides (Saytzeff's rule), Reactions: cis-addition (alk. KMnO_4) and trans-addition (bromine), Addition of HX (Markownikoff's and antiMarkownikoff's addition), Hydration, Ozonolysis, oxymecuration-demercuration, Hydroboration-oxidation.

Alkynes: (3 classes each of 60 minutes duration) (Upto 5 Carbons)

Preparation: Acetylene from CaC_2 and conversion into higher alkynes, by dehalogenation of tetra halides and dehydrohalogenation of vicinal-dihalides.

Reactions: Formation of metal acetylides, addition of bromine and alkaline KMnO_4 , ozonolysis and oxidation with hot alkaline KMnO_4 .

UNIT VII: Aromatic hydrocarbons: (5 classes each of 60 minutes duration)

Preparation of benzene: from phenol, by decarboxylation, from acetylene, from benzene sulphonic acid. Reactions of benzene: Electrophilic substitution: nitration, halogenation and sulphonation. Friedel-Craft's reaction (alkylation and acylation). Side chain oxidation of alkyl benzenes (upto 4 carbons on benzene)

Reference Books:

1. J. C. Kotz, P. M. Treichel & J. R. Townsend: General Chemistry, Cengage Lening India Pvt. Ltd., New Delhi (2009)
2. Lee, J. D. Concise Inorganic Chemistry, Wiley, 5th Edn.
3. Douglas, B.E., McDaniel, D.H., Alexander J.J., Concepts & Models of Inorganic Chemistry, (Third Edition) John Wiley & Sons, 1999.
4. Atkins, P. W. and De Paula, J. Physical Chemistry, Tenth Edition, Oxford University Press, 2014.
5. Douglas, B.E, Mc Daniel, D.H. & Alexander, J.J. Concepts & Models of Inorganic Chemistry 3rd Ed., John Wiley Sons, N.Y. 1994.
6. Peter Sykes, A Guide Book to Mechanism in Organic Chemistry, Longman.
7. C. K. Ingold, Structure and Mechanism in Organic Chemistry, Cornell University Press.
8. R. T. Morrison and R. N. Boyd, Organic Chemistry, Prentice-Hall.
9. H. O. House, Modern Organic Reactions, Benjamin.
10. O. C. Norman and J. M. Coxon, Principles of Organic Synthesis, Blackle Academic & Professional.
11. Ali, Hashmat, Reaction Mechanism in Organic Chemistry ,S Chand

MN-CHE-1A: MINOR PRACTICAL-1

Credit: Theory-01, 30 Hours, Full Marks=25, Pass Marks= 10

End Semester Examination (ESE):

There will be one Practical Examination of 3 Hours duration. Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 15 marks

Practical record notebook = 05 marks

Viva-voce = 05 marks

Section A: Physical

Thermochemistry

1. Determination of heat capacity of calorimeter.
2. Determination of enthalpy of neutralization of hydrochloric acid with sodium hydroxide.
3. Determination of integral enthalpy of solution of salts (KNO_3 , NH_4Cl).
4. Determination of enthalpy of hydration of copper sulphate.

Section B: Inorganic Chemistry - Volumetric Analysis

1. Acid-Base Titrations

- a. Estimation of oxalic acid present in the supplied sample.
- b. Estimation of sodium hydroxide present in given sample.
- c. Estimation of amount of acetic acid in vinegar solution.
- d. Estimation of carbonate and hydroxide present together in mixture.
- e. Estimation of carbonate and bicarbonate present together in a mixture.
- f. Estimation of free alkali present in different soaps/detergents.

2. Oxidation-Reduction Titrimetry

- a. Estimation of Fe(II) in supplied solution using standardized KMnO_4 solution.
- b. Estimation of oxalic acid using standardized KMnO_4 solution.
- c. Estimation of percentage of Fe(II) in Iron fillings with standard $\text{K}_2\text{Cr}_2\text{O}_7$

Section C: Organic Chemistry

1. Purification of organic compounds by crystallization (from water and alcohol) and distillation.
2. Criteria of Purity: Determination of melting and boiling points.
3. Recrystallisation, determination of melting point and calculation of quantitative yields to be done.
 - a. Benzoylation of amines/phenols
 - b. Oxime and 2,4 dinitrophenyl hydrazone of aldehyde/ketone

Reference Books:

1. Vogel's Qualitative Inorganic Analysis, A.I. Vogel, Prentice Hall, 7th Edition.
2. F. G. Mann & B. C. Saunders, Practical Organic Chemistry, Orient Longman (1960).
3. B.D. Khosla, Senior Practical Physical Chemistry, R. Chand & Co.
4. S. M. Khopkar, Environmental Pollution Analysis: Wiley Eastern Ltd, New Delhi.

SEMESTER-III
Minor Course

MN-CHE-2B(THEORY)-: CHEMICAL EQUILIBRIA & FUNCTIONAL GROUPS

Credit: Theory-03, 45 Hours, Full Marks=75, Pass Marks= 30

Marks: 15 (5 Att'd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs)=75

Pass Marks: Th (SIE + ESE) = 30

Instruction to Question Setter for

Semester Internal Examination (SIE 10+5=15 marks):

There will be two group of questions.

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 10 Mark. Question No.1 will be very short answer type in Group A consisting of five questions of 1 mark each. Group B will contain descriptive type two questions of five marks each, out of which any one to answer. (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 60 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No.2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type five questions of fifteen marks each, out of which any three are to answer.

Note: There may be subdivisions in the questions of group B.

Section A: Physical Chemistry

UNIT I: Equilibrium: (15 classes each of 60 minutes duration)

Chemical Equilibria: Le Chatelier's principle. Relationships between K_p , K_c and K_x for reactions involving ideal gases. Free energy change in a chemical reaction. Thermodynamic derivation of the law of chemical equilibrium. Distinction between ΔG° and ΔG ,

Ionic Equilibria: Strong, moderate and weak electrolytes, degree of ionization, factors affecting degree of ionization, ionization constant and ionic product of water. Ionization of weak acids and bases, pH scale, common ion effect. Salt hydrolysis-calculation of hydrolysis constant, degree of hydrolysis and pH for different salts. Buffer solutions. Solubility and solubility product of sparingly soluble salts – applications of solubility product principle.

Section B: Inorganic Chemistry (8 classes each of 60 minutes duration)

MO Approach: Rules for the LCAO method, bonding and antibonding MOs and their characteristics for s-s, sp and p-p combinations of atomic orbitals, nonbonding combination of orbitals, MO treatment of homonuclear diatomic molecules of 1st and 2nd periods (including idea of s-p mixing) and heteronuclear diatomic molecules such as CO, NO and NO. Comparison of VB and MO approaches.

Section C: Organic Chemistry

UNIT II: Alkyl and Aryl Halides

Alkyl Halides (Upto 5 Carbons) (5 classes each of 60 minutes duration)

Types of Nucleophilic Substitution (SN^1 , SN^2 and SN_i) reactions. Preparation: from alkenes and alcohols. Reactions: hydrolysis, nitrite & nitro formation, nitrile & isonitrile formation.

Williamson's ether synthesis: Elimination vs substitution.

Aryl Halides (3 classes each of 60 minutes duration)

Preparation: (Chloro, bromo and iodo-benzene case): from phenol, Sandmeyer & Gattermann reactions. Reactions (Chlorobenzene): Aromatic nucleophilic substitution (replacement by -OH group) and effect of nitro substituent. Benzyne Mechanism: KNH_2/NH_3 (or $NaNH_2/NH_3$).

Reactivity and Relative strength of C-Halogen bond in alkyl, allyl, benzyl, vinyl and aryl halides.

Alcohols: (4 classes each of 60 minutes duration)

Preparation: Preparation of 1° , 2° and 3° alcohols: using Grignard reagent, Ester hydrolysis, Reduction of aldehydes, ketones, carboxylic acid and esters.

Reactions: With sodium, HX (Lucas test), esterification, oxidation (with PCC, alk. KMnO_4 , acidic dichromate, conc. HNO_3). Oppeneauer oxidation Diols: (Upto 6 Carbons) oxidation of diols. Pinacol-Pinacolone rearrangement.

Phenols: (3 classes each of 60 minutes duration)

Preparation: Cumene hydroperoxide method, from diazonium salts.

Reactions: Electrophilic substitution: Nitration, halogenation and sulphonation. Reimer- Tiemann Reaction, Gattermann-Koch Reaction, Houben–Hoesch Condensation, Schotten – Baumann Reaction.

Ethers (aliphatic and aromatic): (2 classes each of 60 minutes duration)

Cleavage of ethers with HI.

Aldehydes and ketones (aliphatic and aromatic): (5 classes each of 60 minutes duration)

(Formaldehyde, acetaldehyde, acetone and benzaldehyde) Preparation: from acid chlorides and from nitriles.

Reactions– Reaction with HCN, ROH, NaHSO_3 , $\text{NH}_2\text{-G}$ derivatives. Iodoform test. Aldol Condensation, Cannizzaro's reaction, Wittig reaction, Benzoin condensation. Clemensen reduction and Wolff Kishner reduction. Meerwein-Ponndorf Verley reduction.

Reference Books:

1. T. W. Graham Solomons: Organic Chemistry, John Wiley and Sons.
2. Peter Sykes: A Guide Book to Mechanism in Organic Chemistry, Orient Longman.
3. I.L. Finar: Organic Chemistry (Vol. I & II), E. L. B. S.
4. R. T. Morrison & R. N. Boyd: Organic Chemistry, Prentice Hall.
5. Arun Bahl and B. S. Bahl: Advanced Organic Chemistry, S. Chand.
6. G. M. Barrow: Physical Chemistry Tata McGraw-Hill (2007).
7. G. W. Castellan: Physical Chemistry 4th Edn. Narosa (2004).
8. J. C. Kotz, P. M. Treichel & J. R. Townsend: General Chemistry Cengage Learning India Pvt. Ltd., New Delhi (2009).
9. B. H. Mahan: University Chemistry 3rd Ed. Narosa (1998).
10. R. H. Petrucci: General Chemistry 5th Ed. Macmillan Publishing Co.: New York (1985).
11. Ali, Hashmat, Reaction Mechanism in Organic Chemistry, S Chand

MN-CHE-2B: MINOR PRACTICAL-II

Credit: Theory-01, 30 Hours, Full Marks=25, Pass Marks= 10

End Semester Examination (ESE):

There will be one Practical Examination of 3 Hours duration. Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 15 marks

Practical record notebook = 05 marks

Viva-voce = 05 marks

Section A: Physical Chemistry

Ionic equilibria pH measurements

1. Measurement of pH of different solutions like aerated drinks, fruit juices, shampoos and soaps (use dilute solutions of soaps and shampoos to prevent damage to the glass electrode) using pH-meter.
2. Determine the pH of the given aerated drinks fruit juices, shampoos and soaps.
3. Preparation of buffer solutions:
 - a. Sodium acetate-acetic acid
 - b. Ammonium chloride-ammonium hydroxide

Section B: Organic Chemistry

1. Detection of hetero elements in organic compounds.
2. Functional group tests for alcohols, phenols, carbonyl and carboxylic acid group.
3. Purification of organic compounds by crystallization (from water and alcohol) and distillation.
4. Criteria of Purity: Determination of melting and boiling points.

5. Preparations: Mechanism of various reactions involved to be discussed.
6. Recrystallisation, determination of melting point and calculation of quantitative yields to be done.
A. Oxime of aldehyde/ketone
b. 2,4 dinitrophenylhydrazones of aldehyde/ketone
7. Analysis of soaps and detergents.
8. Preparation of Nylon-6, Nylon-66
9. Preparation of face cream
10. Vitamin-C preparation.

Reference Books

1. B.D. Khosla, Senior Practical Physical Chemistry, R. Chand & Co.
2. A.I. Vogel: Textbook of Practical Organic Chemistry, 5th edition, Prentice-Hall.
3. F. G. Mann & B. C. Saunders, Practical Organic Chemistry, Orient Longman (1960).
4. Waites M.J. (2008). Industrial Microbiology: An Introduction, 7th Edition, Blackwell Science, London, UK.
5. Prescott S.C., Dunn C.G., Reed G. (1982). Prescott & Dunn's Industrial Microbiology, 4th Edition, AVI Pub. Co., USA.
6. Reed G. (2004). Prescott & Dunn's industrial microbiology, 4th Edition, AVI Pub. Co., USA.
7. JR Casida L.E. (2015). Industrial Microbiology, 3rd Edition, New Age International (P) Limited Publishers, New Delhi, India.
8. Waites M.J., Morgan N.L., Rockey J.S. and Highton G. (2001) Industrial Microbiology: An Introduction. 1st Edition, Blackwell Science, London, UK.
9. Pelczar M.J., Chan E.C.S. and Krieg N.R. (2003) Microbiology. 5th Edition, Tata McGraw-Hill Publishing Company Limited, New Delhi.

SEMESTER-V
Minor Course

MN-CHE-3C(THEORY)-: CHEMISTRY OF s- & p-BLOCK ELEMENTS AND STATES OF MATTER

Credit: Theory-03, 45 Hours, Full Marks=75, Pass Marks= 30

Marks: 15 (5 Attd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs) =75

Pass Marks: Th (SIE + ESE) = 30

Instruction to Question Setter for

Semester Internal Examination (SIE 10+5=15 marks):

There will be two group of questions.

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 10 Mark. Question No.1 will be very short answer type in Group A consisting of five questions of 1 mark each. Group B will contain descriptive type two questions of five marks each, out of which any one to answer. (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 60 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No.2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type five questions of fifteen marks each, out of which any three are to answer.

Note: There may be subdivisions in the questions of group B.

SECTION-A: Inorganic Chemistry

UNIT I: General Principles of Metallurgy: (5 Lectures)

Chief modes of occurrence of metals based on standard electrode potentials. Ellingham diagrams for reduction of metal oxides using carbon as reducing agent.

Hydrometallurgy, Methods of purification of metals (Al, Pb, Ti, Fe, Cu, Ni, Zn): electrolytic, oxidative refining, Kroll process, Parting process, van Arkel-de Boer process and Mond's process.

UNIT II: s- and p-Block Elements: (5 Lectures)

Periodicity in s- and p-block elements with respect to electronic configuration, atomic and ionic size, ionization enthalpy, electronegativity (Pauling, Mulliken, and Alfred-Rochow scales).

Allotropy in C, S, and P.

Oxidation states with reference to elements in unusual and rare oxidation states like carbides and nitrides), inert pair effect, diagonal relationship and anomalous behaviour of first member of each group.

UNIT III: Compounds of s- and p-Block Elements: (10 Lectures)

Hydrides and their classification (ionic, covalent and interstitial), structure and properties with respect to stability of hydrides of p- block elements. Concept of multicentre bonding (diborane). Structure, bonding and their important properties like oxidation/reduction, acidic/basic nature of the following compounds and their applications in industrial, organic and environmental chemistry. Hydrides of nitrogen (NH₃, N₂H₄, N₃H, NH₂OH) Oxoacids of P, S and Cl. Halides and oxohalides: PCl₃, PCl₅, SOCl₂ and SO₂Cl₂

Section B: Physical Chemistry

UNIT IV: Kinetic Theory of Gases: (15 Lectures)

Postulates of Kinetic Theory of Gases and derivation of the kinetic gas equation. Most probable, average and root mean square velocities (no derivation). Collision number, collision frequency, collision diameter and mean free path of molecules. Maxwell Boltzmann distribution laws of molecular velocities and molecular energies (graphic representation – derivation not required) and their importance.

Deviation of real gases from ideal behaviour, compressibility factor, causes of deviation. Van der Waals equation of state for real gases. Boyle temperature (derivation not required).

Critical phenomena, critical constants and their calculation from van der Waals equation.

Andrews isotherms of CO₂. Viscosity of gases and effect of temperature and pressure on coefficient of viscosity (qualitative treatment only).

UNIT V: Liquids: (4 Lectures)

Surface tension and its determination using stalagmometer. Viscosity of a liquid and determination of coefficient of viscosity using Ostwald viscometer. Effect of temperature on surface tension and coefficient of viscosity of a liquid (qualitative treatment only)

UNIT VI: Solids (6 Lectures)

Forms of solids. Symmetry elements, unit cells, crystal systems, Bravais lattice types and identification of lattice planes. Laws of Crystallography – Law of constancy of interfacial angles, Law of rational indices. Miller indices. X-Ray diffraction by crystals, Bragg's law. Structures of NaCl, KCl and CsCl (qualitative treatment only). Defects in crystals. Glasses and liquid crystals.

Reference Books:

1. G. M. Barrow: Physical Chemistry Tata McGraw-Hill (2007).
2. G. W. Castellan: Physical Chemistry 4th Edn. Narosa (2004).
3. J. C. Kotz, P. M. Treichel & J. R. Townsend: General Chemistry Cengage Learning India Pvt. Ltd., New Delhi (2009).
4. B. H. Mahan: University Chemistry 3rd Ed. Narosa (1998).
5. R. H. Petrucci: General Chemistry 5th Ed. Macmillan Publishing Co.: New York (1985).
6. J. D. Lee: A New Concise Inorganic Chemistry, E.L.B.S.
7. F.A. Cotton & G. Wilkinson: Basic Inorganic Chemistry, John Wiley.
8. D. F. Shriver and P. W. Atkins: Inorganic Chemistry, Oxford University Press.
9. Gary Wulfsberg: Inorganic Chemistry, Viva Books Pvt. Ltd.

MN-CHE-3C-MINOR PRACTICAL-III

Credit: Theory-01, 30 Hours, Full Marks=25, Pass Marks= 10

End Semester Examination (ESE):

There will be one Practical Examination of 3 Hours duration. Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 15 marks

Practical record notebook = 05 marks

Viva-voce = 05 marks

Section A: Inorganic Chemistry

Qualitative semi micro analysis

1. Semi-micro qualitative analysis using H₂S of mixtures- not more than four ionic species (two anions and two cations and excluding insoluble salts) out of the following:
Cations: NH₄⁺, Pb²⁺, Ag⁺, Bi³⁺, Cu²⁺, Cd²⁺, Sn²⁺, Fe³⁺, Al³⁺, Co²⁺, Cr³⁺, Ni²⁺, Mn²⁺, Zn²⁺, Ba²⁺, Sr²⁺, Ca²⁺, K⁺
Anions : CO₃²⁻, S²⁻, SO₃²⁻, S₂O₃²⁻, NO₃⁻, CH₃COO⁻, Cl⁻, Br⁻, I⁻, NO₃⁻, SO₄²⁻, PO₄³⁻, BO₃³⁻, C₂O₄²⁻, F⁻ (Spot tests should be carried out wherever feasible)

Section B: Physical Chemistry

(I) Surface tension measurement (use of organic solvents excluded).

- a) Determination of the surface tension of a liquid or a dilute solution using a stalagmometer.
- b) Study of the variation of surface tension of a detergent solution with concentration.

(II) Viscosity measurement (use of organic solvents excluded).

- a) Determination of the relative and absolute viscosity of a liquid or dilute solution using an Ostwald's viscometer.
- b) Study of the variation of viscosity of an aqueous solution with concentration of solute.

(III) Chemical Kinetics

Study the kinetics of the following reactions.

- a. Initial rate method: Iodide-persulphate reaction
- b. Integrated rate method:

- c. Acid hydrolysis of methyl acetate with hydrochloric acid.
- d. Saponification of ethyl acetate.
- e. Compare the strengths of HCl and H₂SO₄ by studying kinetics of hydrolysis of methyl acetate

Reference Books

1. A.I. Vogel, Qualitative Inorganic Analysis, Prentice Hall, 7th Edn.
2. A.I. Vogel, Quantitative Chemical Analysis, Prentice Hall, 6th Edn.
3. B.D. Khosla, Senior Practical Physical Chemistry, R. Chand & Co.

SEMESTER-VII
Minor Course

MN-CHE-4D(THEORY)-: CHEMISTRY OF d- & f-BLOCK ELEMENTS & MOLECULES OF LIFE

Credit: Theory-03, 45 Hours, Full Marks=75, Pass Marks= 30

Marks: 15 (5 Att'd. + 10 SIE: 1Hr) + 60 (ESE: 3Hrs)=75

Pass Marks: Th (SIE + ESE) = 30

Instruction to Question Setter for

Semester Internal Examination (SIE 10+5=15 marks):

There will be two group of questions.

The Semester Internal Examination shall have two components. (a) One Semester Internal Examination Written Test (SIE) of 10 Mark. Question No.1 will be very short answer type in Group A consisting of five questions of 1 mark each. Group B will contain descriptive type two questions of five marks each, out of which any one to answer. (b) Class Attendance Score (CAS) including the behaviour of the student towards teachers and other students of the College of 5 marks.

End Semester Examination (ESE 60 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No.2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type five questions of fifteen marks each, out of which any three are to answer.

Note: There may be subdivisions in the questions of group B.

Section A: Inorganic Chemistry

UNIT I: Transition Elements (3d series) (6 classes each of 60 minutes duration)

General group trends with special reference to electronic configuration, variable valency, colour, magnetic and catalytic properties, ability to form complexes and stability of various oxidation states.

UNIT II: Lanthanides and Actinides: (5 classes each of 60 minutes duration)

Electronic configuration, oxidation states, colour, spectra and magnetic behaviour of lanthanides and actinides. Lanthanide contraction, separation of lanthanides (ion-exchange method only).

UNIT III: Coordination Chemistry (5 classes each of 60 minutes duration)

Valence Bond Theory (VBT): Inner and outer orbital complexes of Cr, Fe, Co, Ni and Cu (coordination numbers 4 and 6). Structural and stereoisomerism in complexes with coordination numbers 4 and 6. Drawbacks of VBT. IUPAC system of nomenclature.

UNIT IV: Crystal Field Theory (4 classes each of 60 minutes duration)

Crystal field effect, octahedral symmetry. Crystal field stabilization energy (CFSE), Crystal field effects for weak and strong fields. Tetrahedral symmetry. Factors affecting the magnitude of D. Spectrochemical series. Comparison of CFSE for Oh and Td complexes.

Section B: Organic Chemistry

UNIT V: Carbohydrates (8 classes each of 60 minutes duration)

Classification of carbohydrates, reducing and non-reducing sugars, General properties of Glucose and Fructose, their open chain structure. Epimers, mutarotation and anomers. Determination of configuration of Glucose (Fischer proof). Cyclic structure of glucose. Haworth projections. Cyclic structure of fructose. Linkage between monosachharides, structure of disacharrides (sucrose, maltose, lactose) and polysacharrides (starch and cellulose) excluding their structure elucidation.

UNIT VI: Amino Acids, Peptides and Proteins (8 classes each of 60 minutes duration)

Classification of Amino Acids, Zwitterion structure and Isoelectric point

Overview of Primary, Secondary, Tertiary and Quaternary structure of proteins.

Determination of primary structure of peptides, determination of N-terminal amino acid (by DNFB and Edman method) and C-terminal amino acid (by thiohydantoin and with carboxypeptidase enzyme). Synthesis of simple peptides (upto dipeptides) by N-protection (t-butyloxycarbonyl and phthaloyl) & Cactivating groups and Merrifield solid phase synthesis.

UNIT VII: Enzymes and correlation with drug action (6 classes each of 60 minutes duration)

Mechanism of enzyme action, factors affecting enzyme action, Coenzymes and cofactors and their role in biological reactions, Specificity of enzyme action (Including stereospecificity).

Enzyme inhibitors and their importance, phenomenon of inhibition (Competitive and Non-competitive inhibition including allosteric inhibition).

Drug action-receptor theory. Structure –activity relationships of drug molecules, binding role of –OH group, NH₂ group, double bond and aromatic ring,

UNIT VIII: Lipids (3 classes each of 60 minutes duration)

Introduction to lipids, classification. Oils and fats: Common fatty acids present in oils and fats, Omega fatty acids, Trans fats, Hydrogenation, Saponification value, Iodine number.

Biological importance of triglycerides, phospholipids, glycolipids, and steroids (cholesterol).

Reference Books:

1. J. C. Kotz, P. M. Treichel & J. R. Townsend: General Chemistry, Cengage Learning India Pvt. Ltd., New Delhi (2009).
2. B. H. Mahan: University Chemistry 3rd Ed. Narosa (1998).
3. R. H. Petrucci: General Chemistry 5th Ed. Macmillan Publishing Co.: New York (1985).
4. J. D. Lee: A New Concise Inorganic Chemistry, E.L.B.S.
5. F.A. Cotton & G. Wilkinson: Basic Inorganic Chemistry, John Wiley.
6. Gary Wulfsberg: Inorganic Chemistry, Viva Books Pvt. Ltd.
7. Morrison, R. T. & Boyd, R. N. Organic Chemistry, Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
8. Finar, I. L. Organic Chemistry (Volume 1), Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
9. Finar, I. L. Organic Chemistry (Volume 2), Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
10. Nelson, D. L. & Cox, M. M. Lehninger's Principles of Biochemistry 7th Ed., W. H. Freeman.
11. Berg, J. M., Tymoczko, J. L. & Stryer, L. Biochemistry 7th Ed., W. H. Freeman.
12. Ali, Hashmat, Reaction Mechanism in Organic Chemistry, S Chand

MN-CHE-4D: MINOR PRACTICAL-IV

Credit: Theory-01, 30 Hours, Full Marks=25, Pass Marks= 10

End Semester Examination (ESE):

There will be one Practical Examination of 3 Hours duration. Evaluation of Practical Examination may be as per the following guidelines:

One Experiment = 15 marks

Practical record notebook = 05 marks

Viva-voce = 05 marks

Section A: Inorganic Chemistry

1. Estimation of the amount of nickel present in a given solution as bis(dimethylglyoximate) nickel (II) or aluminium as oxinate in a given solution gravimetrically.
2. Estimation of (i) Mg or (ii) Zn by complexometric titrations using EDTA.
3. Estimation of total hardness of a given sample of water by complexometric titration.

Section B: Organic Chemistry

1. Separation of amino acids by paper chromatography
2. To determine the concentration of glycine solution by formylation method.
3. Study of titration curve of glycine
4. To determine the saponification value of an oil/fat.
5. To determine the iodine value of an oil/fat
6. Differentiate between a reducing/ nonreducing sugar.
7. To synthesise aspirin by acetylation of salicylic acid and compare it with the ingredient of an aspirin tablet by TLC.

Reference Books

1. A.I. Vogel, Qualitative Inorganic Analysis, Prentice Hall, 7th Edn.
2. A.I. Vogel, Quantitative Chemical Analysis, Prentice Hall, 6th Edn.
3. Vogel's Textbook of Practical Organic Chemistry, ELBS.
4. Ahluwalia, V.K. & Aggarwal, R. Comprehensive Practical Organic Chemistry, Universities Press.

MULTIDISCIPLINARY COURSE IN CHEMISTRY
FOR SEMESTER-I/II/III

Subject Code: MDC-CHE

Credit: Theory-03, Full Marks=75, Pass Marks= 30, Lectures:45

Instruction to Question Setter for

End Semester Examination (ESE 75 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type six questions of fifteen marks each, out of which any four are to answer.

Note: There may be subdivisions in the questions of group B.

Unit-I: Food & Nutrition (12 Lectures)

Basic concept on Food, Nutrition and Nutrients. Classification of Food, Classification of Nutrients. Carbohydrates-Definition, sources, classification, and properties Fatty acids-composition, properties, types. Lipids -Definition, sources, Classification & Properties, daily requirements, function Role & nutritional significances of PUFA, MUFA, SFA, W-3 fatty acid. Proteins- Definition, Sources, classification & properties, daily requirements, functions. Assessment of Protein quality (BV, PER, NPU). Vitamins- Definition, sources, classification, diseases caused by deficiency of Vitamins. Minerals - Definition, sources, classification, diseases caused by deficiency of minerals.

Unit-II Chemicals in Food (6 Lectures)

Introductory idea of Food colours, Flavours and sweeteners, Fat emulsifiers and stabilizing agents, Flour improvers - antistaling agents and bleaches, Antioxidants, Nutritional supplements such as minerals, vitamins and amino acids.

Unit-III Food Preservation (6 Lectures)

Food preservation: definition, objectives and principles of food preservation. Different methods of food preservation. Preserved Products: Jelly, Marmalade, Sauces, Pickles, Squashes, Syrups- Definitions, types, composition, storage, uses and nutritional aspects, Food Standards: ISI, Agmark, FPO, MPO, PFA, FSSAI.

Unit-IV Agrochemicals (6 Lectures)

Herbicides-Definition, Classification, properties, uses and health hazards, Fungicides Definition, Classification, properties, uses and health hazards, Insecticides- Definition, Classification, properties, uses and health hazards, traditional pesticides.

Unit-V Drugs and Medicines (6 Lectures)

Drugs- Definition, Classification, Antacids, Antihistamines, Tranquilizers, analgesics, antidepressant drugs, Antimicrobials, Antibiotics, Antiseptics and disinfectants, medicinal uses of common herbs and plants.

Unit-VI Chemistry of Materials (9 Lectures)

Soaps and Detergents – Definition, Classification & their action, Biofuels – Definition, Classification, production of biofuels and its utility as alternative fuel source, Fibers: Definition, Classification and their uses, natural fibers, cotton, wool, silk, rayon, artificial fibers, polyamides, acrylic acid, PVC, PVA; Examples of natural biodegradable polymers synthetic biodegradable polymers. Use of polymeric materials in daily life.

Reference:

1. SrilakshmiB(2017): Nutrition Science,6th Multicolour Ed. New Age International (P) Ltd.
2. RodayS(2012): Food Science and Nutrition, 2nd Ed. Oxford University Press.
3. Subalakshmi, G and Udipi, SA(2006):Food processing and preservation, 1st Ed. New Age International (P)Ltd.
4. Agrochemicals preparation and mode of action, Cremlyn, R.J.W.
5. Insecticide, Action and Metabolism, O'Brien, R.D., Academic Press, New York and London.
6. Chemical Pesticides: Mode of Action and Toxicology, Stenersen, J., CRC, 2004
7. Indian pharmacopoeia

INSTRUCTIONS TO QUESTION SETTER

SEMESTER INTERNAL EXAMINATION (SIE):

There will be Only One Semester Internal Examination in Major, Minor and Research Courses, which will be organized at college/institution level. However, Only One End semester evaluation in other courses will be done either at College/ Institution or University level depending upon the nature of course in the curriculum.

A. (SIE 10+5=15 marks):

There will be two group of questions. Question No.1 will be very short answer type in Group A consisting of five questions of 1 mark each. Group B will contain descriptive type two questions of five marks each, out of which any one to answer.

The Semester Internal Examination shall have two components. (a) One Semester Internal Assessment Test (SIA) of 10 Marks, (b) Class Attendance Score (CAS) of 5 marks.

B. (SIE 20+5=25 marks):

There will be two group of questions. Group A is compulsory which will contain two questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No.2 will be short answer type of 5 marks. Group B will contain descriptive type two questions of ten marks each, out of which any one to answer.

The Semester Internal Examination shall have two components. (a) One Semester Internal Assessment Test (SIA) of 20 Marks, (b) Class Attendance Score (CAS) of 5 marks.

Conversion of Attendance into score may be as follows:

Attendance Upto 45%, 1 mark; 45<Attd.<55, 2 marks; 55<Attd.<65, 3 marks; 65<Attd.<75, 4 marks; 75<Attd, 5 marks.

END SEMESTER UNIVERSITY EXAMINATION (ESE):

A. (ESE 60 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No.2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type five questions of fifteen marks each, out of which any three are to answer.

B. (ESE 75 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type six questions of fifteen marks each, out of which any four are to answer.

C. (ESE 100 marks):

There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of ten questions of 1 mark each. Question No. 2 & 3 will be short answer type of 5 marks. Group B will contain descriptive type six questions of twenty marks each, out of which any four are to answer.

FORMAT OF QUESTION PAPER FOR SEMESTER INTERNAL EXAMINATION

Question format for 10 Marks:

F.M. =10	Subject/ Code	Exam Year
Time=1Hr.		
General Instructions:		
i. Group A carries very short answer type compulsory questions. ii. Answer 1 out of 2 subjective/ descriptive questions given in Group B . iii. Answer in your own words as far as practicable. iv. Answer all sub parts of a question at one place. v. Numbers in right indicate full marks of the question.		
<u>Group A</u>		
i.		[5x1=5]
ii.		
iii.		
iv.		
v.		
<u>Group B</u>		
2.		[5]
3.		[5]
Note: There may be subdivisions in each question asked in Theory Examination.		

Question format for 20 Marks:

F.M. =20	Subject/ Code	Exam Year
Time=1Hr.		
General Instructions:		
i. Group A carries very short answer type compulsory questions. ii. Answer 1 out of 2 subjective/ descriptive questions given in Group B . iii. Answer in your own words as far as practicable. iv. Answer all sub parts of a question at one place. v. Numbers in right indicate full marks of the question.		
<u>Group A</u>		
1.		[5x1=5]
i.		
ii.		
iii.		
iv.		
v.		
2.		[5]
<u>Group B</u>		
3.		[10]
4.		[10]
Note: There may be subdivisions in each question asked in Theory Examination.		

FORMAT OF QUESTION PAPER FOR END SEMESTER UNIVERSITY EXAMINATION

Question format for 50 Marks:

F.M. =50	Subject/ Code	Exam Year
Time=3Hrs.		
General Instructions:		
i. Group A carries very short answer type compulsory questions. ii. Answer 3 out of 5 subjective/ descriptive questions given in Group B . iii. Answer in your own words as far as practicable. iv. Answer all sub parts of a question at one place. v. Numbers in right indicate full marks of the question.		
<u>Group A</u>		[5x1=5]
i.		
ii.		
iii.		
iv.		
v.		
<u>Group B</u>		
2.		[15]
3.		[15]
4.		[15]
5.		[15]
6.		[15]
Note: There may be subdivisions in each question asked in Theory Examination.		

Question format for 60 Marks:

F.M. =60	Subject/ Code	Exam Year
Time=3Hrs.		
General Instructions:		
i. Group A carries very short answer type compulsory questions. ii. Answer 3 out of 5 subjective/ descriptive questions given in Group B . iii. Answer in your own words as far as practicable. iv. Answer all sub parts of a question at one place. v. Numbers in right indicate full marks of the question.		
<u>Group A</u>		[5x1=5]
1.		
i.		
ii.		
iii.		
iv.		
v.		
2.		[5]
3.		[5]
<u>Group B</u>		
4.		[15]
5.		[15]
6.		[15]
7.		[15]
8.		[15]
Note: There may be subdivisions in each question asked in Theory Examination.		

Question format for 75 Marks:

F.M. = 75	Subject/ Code	Exam Year
Time=3Hrs.		
General Instructions:		
i. Group A carries very short answer type compulsory questions. ii. Answer 4 out of 6 subjective/ descriptive questions given in Group B. iii. Answer in your own words as far as practicable. iv. Answer all sub parts of a question at one place. v. Numbers in right indicate full marks of the question.		
<u>Group A</u>		
1.		[5x1=5]
i.		
ii.		
iii.		
iv.		
v.		
2.		[5]
3.		[5]
<u>Group B</u>		
4.		[15]
5.		[15]
6.		[15]
7.		[15]
8.		[15]
9.		[15]
Note: There may be subdivisions in each question asked in Theory Examination.		

Question format for 100 Marks:

F.M. = 100	Subject/ Code	Exam Year
Time=3Hrs.		
General Instructions:		
i. Group A carries very short answer type compulsory questions. ii. Answer 4 out of 6 subjective/ descriptive questions given in Group B. iii. Answer in your own words as far as practicable. iv. Answer all sub parts of a question at one place. v. Numbers in right indicate full marks of the question.		
<u>Group A</u>		
i.		[10x1=10]
ii.		
iii.		
iv.		
v.		
vi.		
vii.		
viii.		
ix.		
x.		
2.		[5]
3.		[5]
<u>Group B</u>		
4.		[20]
5.		[20]
6.		[20]
7.		[20]
8.		[20]
9.		[20]
Note: There may be subdivisions in each question asked in Theory Examination.		